MENTAL HEALTH SUPPORT PLATFORM

Software Requirements & System Design Report

Project Type: Web Application (Full Stack)

Duration: 12 Weeks

Institution: Kent Institute of Technology Australia

Industry Sponsor: Skillup Labs

Team: Mel23 Tech Solution

Date: April 2026

Team Members

Sanjana Tanwar (K240225) — Team Leader / Project Manager

Jubair Zaman Dipto (K240568) — Full-Stack / Tech Lead

S M Nalid Maola (K231528) — Backend Developer

Nabil Ashrafi (K231720) — Frontend Developer

Md Al Amin Sikder (K240094) — QA Engineer / Tester

Tranpreet Kaur Kumar (K241957) — UI/UX Designer

Academic Supervisor: Mr. Syed Altaf

Industry Sponsor Contact: Nabin Singh (wil@skilluplabs.com.au)

Table of Contents

1. Introduction

1.1 Purpose of the Project

1.2 Objective of the Project

1.3 Project Scope and Limitations

1.4 Assumptions

1.5 Company Profile

1.6 Organisation Chart

1.7 Roles and Responsibilities

1.8 Key Deliverables

1.9 Project Budget

1.10 Work Breakdown Structure

1.11 Gantt Chart

2. System Requirements

2.1 Functional Requirements

2.2 Non-Functional Requirements

3. System Architecture

3.1 Software and Hardware Architecture

3.2 Tech Stack (Detailed)

4. System Design

4.1 User Interface Design / Wireframes

5. Use Case Diagrams

6. Context Diagrams (Gane & Sarson Method)

7. Data Flow Diagrams (Gane & Sarson Method)

7.1 DFD Level 1

7.2 DFD Level 2

8. Sequence Diagrams

9. Entity Relationship Diagram & Data Dictionary

10. Class Diagram

11. Data Protection Impact Assessment (DPIA)

12. Test and Implementation Plan

13. Responses on Feedback from Supervisor

References

Appendix

1. Introduction

Mental health plays a huge role in how people feel and function day to day, but getting the right support at the right time is not always easy. A lot of people struggle to keep track of how they are feeling or find trustworthy resources that can actually help. Things like stigma, not knowing where to look, and limited access to services make it even harder for people to reach out.

This document is the Software Requirements Specification (SRS) and System Design report for our Mental Health Support Platform. It is a web-based app that lets users track their mood, browse helpful mental health resources, and use supportive tools — all within a secure and easy-to-use environment.

1.1 Purpose of the Project

The main purpose of this project is to build a safe, easy-to-use web platform where people can look after their mental health. Users will be able to log their mood, write journal entries, and find useful mental health resources — all in one private and secure place.

We want the platform to encourage self-awareness and help people catch warning signs early. Users can record how they feel each day, keep a personal journal, and browse organised materials like articles, wellness tips, and mental health support information. Over time, they will be able to spot patterns in their emotions.

Security is a top priority. The system will use proper authentication and data protection so that users' personal information stays private and safe. On top of that, the platform will present data on a clean, simple dashboard to make it easy to use.

We are following standard Software Development Lifecycle (SDLC) practices throughout this project — from gathering requirements and designing the system, through to building, testing, and deploying a working prototype.

1.2 Objective of the Project

The goal of this project is to design and build a working prototype that helps users keep track of their mental wellbeing and find helpful resources. Here are the main objectives we set out to achieve:

- Look into what users actually need when it comes to mental health tracking and support
- Define system requirements for a secure support platform
- Build a full-stack web app using React for the front end, Node.js for the back end, and PostgreSQL for the database
- Develop mood tracking, journaling, and AI chatbot features
- Set up secure login with JWT tokens, OTP verification, bcrypt password hashing, and AES-256 data encryption
- Give users access to a well-organised library of mental health resources
- Let patients and clinicians message each other securely in real time
- Add appointment booking that syncs with Google Calendar
- Get a working prototype deployed on a cloud hosting platform

1.3 Project Scope and Limitations

Scope:

- Sign-up, login with JWT and OTP, and role-based access so different users see different things
- Daily mood logging on a 1–10 scale with notes and visual trend charts
- Private messaging between patients and clinicians, encrypted with AES-256
- Appointment booking with clash detection and Google Calendar sync
- An AI chatbot that uses NLP and can hand off to a real clinician if its confidence drops below 70%
- A one-tap emergency button that responds in under a second and shows crisis hotline details
- Clinical records that can only be viewed with proper consent
- A resource library where URLs are checked, content is grouped by topic, and views are counted
- An admin dashboard with analytics, audit logs, and compliance reports
- Privacy features like consent management and the option to delete your data

Limitations:

- This is a prototype — it is not meant to diagnose or treat any medical condition
- The AI chatbot gives general support only; it does not replace advice from a qualified professional
- Live video consultations are not included in this version
- All testing will use dummy data, not real patient records
- There is no native mobile app — the site will be responsive so it works on phones and tablets

1.4 Assumptions

- Users will be on a modern browser like Chrome, Firefox, Safari, or Edge
- A stable internet connection is needed for real-time features like chat
- Our team has access to the development tools and cloud services we need
- The third-party APIs we rely on (Google Calendar, OpenAI, Twilio, SendGrid) will stay available while we build
- We are using agile methodology and working in iterative sprints
- Every team member is available for the full 12-week project timeline
- Our academic supervisor and industry sponsor will give us feedback on time
- The PostgreSQL database will be hosted in the cloud with automatic backups turned on

1.5 Company Profile

Mel23 Tech Solution

Mel23 Tech Solution is our student development team, put together at Kent Institute of Technology Australia for the WIL (Work Integrated Learning) capstone project. We work under the guidance of our academic supervisor, Mr. Syed Altaf, and with sponsorship from Skillup Labs.

Industry Sponsor: Skillup Labs

Skillup Labs is the industry partner sponsoring this project. They help student teams build real-world tech products, especially in the HealthTech space.

Attribute	Details
Industry Provider	Skillup Labs

Program Contact	Nabin Singh
Contact Email	wil@skilluplabs.com.au
Academic Supervisor	Mr. Syed Altaf
Institution	Kent Institute of Technology Australia
Project Duration	12 Weeks
Team Size	6 Members

1.6 Organisation Chart

The chart below shows how our project team is structured. It includes the academic supervisor, industry sponsor, team leader, and each team member along with their role.

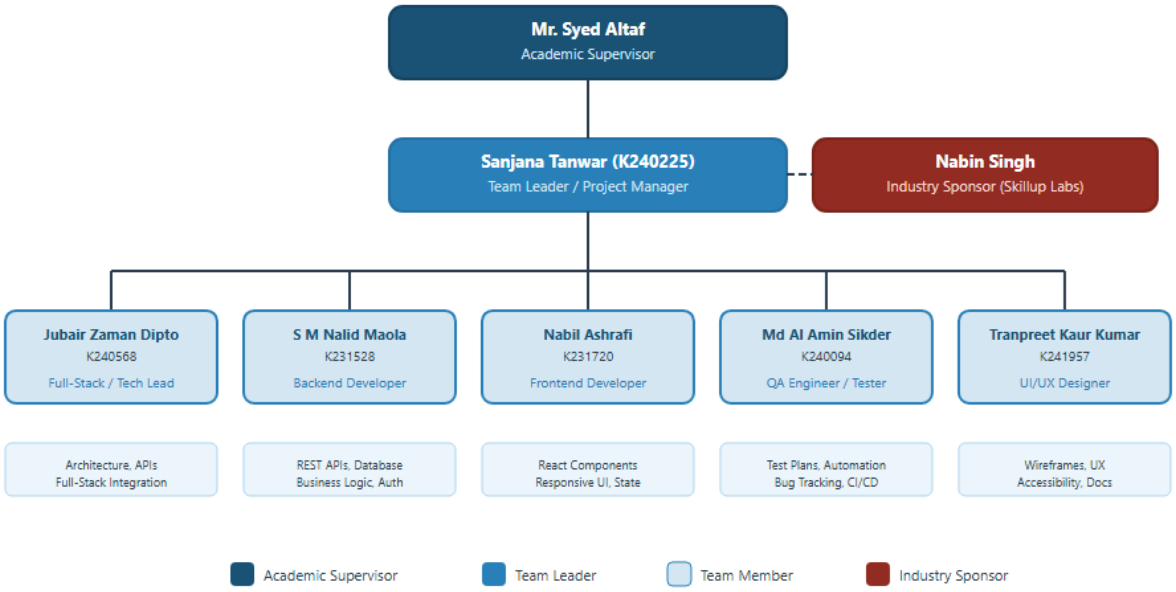

Figure 1.1: Organisation Chart — Mel23 Tech Solution

1.7 Roles and Responsibilities

Name / ID	Role	Responsibilities
Mr. Syed Altaf	Academic Supervisor	Provides academic guidance, reviews deliverables, evaluates team progress
Nabin Singh	Industry Sponsor	Provides industry context, project requirements, and real-world expectations
Sanjana Tanwar K240225	Team Leader / Project Manager	Project coordination, sprint planning, stakeholder communication, progress tracking
Jubair Zaman Dipto K240568	Full-Stack / Tech Lead	System architecture, API design, full-stack integration, code reviews
S M Nalid Maola K231528	Backend Developer	REST API development, database design, business logic, authentication
Nabil Ashrafi K231720	Frontend Developer	React component development, responsive UI, state management
Md Al Amin Sikder K240094	QA Engineer / Tester	Test planning, test automation, bug tracking, CI/CD pipeline

Tranpreet Kaur Kumar K241957	UI/UX Designer	Wireframes, UX research, accessibility compliance, design documentation

1.8 Key Deliverables

ID	Deliverable	Due
D1	Problem Analysis Report	Week 2
D2	Software Requirements Specification (SRS)	Week 3
D3	System Architecture Documentation	Week 4
D4	Database Schema Design (ERD)	Week 5
D5	UI/UX Wireframes (Balsamiq Cloud)	Week 5
D6	Functional Web Application Prototype	Week 9
D7	Source Code Repository (GitHub)	Ongoing
D8	Testing Documentation (Unit, Integration, E2E)	Week 11
D9	Deployment Documentation	Week 11
D10	Final Presentation & Demonstration	Week 12

1.9 Project Budget

Since this is a university capstone project, the budget mainly covers cloud hosting, third-party API costs, and development tools. The table below breaks down the estimated spending.

Item	Estimated Cost	Notes
Cloud Hosting (AWS/Render)	$0 – $50/month	Free tier / student credits
PostgreSQL Database Hosting	$0 – $20/month	Render/Supabase free tier
Domain Name	$10 – $15/year	Optional for demo
OpenAI API (Chatbot)	$10 – $30/month	GPT-3.5/4 usage during development
Twilio SMS API	$0 – $15/month	Trial credits available
SendGrid Email API	$0	Free tier (100 emails/day)
Google Calendar API	$0	Free tier
Balsamiq Cloud (Wireframes)	$0	Educational license
GitHub (Version Control)	$0	Free for students
Development Tools (VS Code etc.)	$0	Free / open-source
Total Estimated Budget	$20 – $130/month	

1.10 Work Breakdown Structure

The Work Breakdown Structure (WBS) splits the project into six main phases: Discovery, Requirements, Design, Development, Testing, and Deployment. Each phase has its own set of tasks and deliverables, as shown in the diagram below.

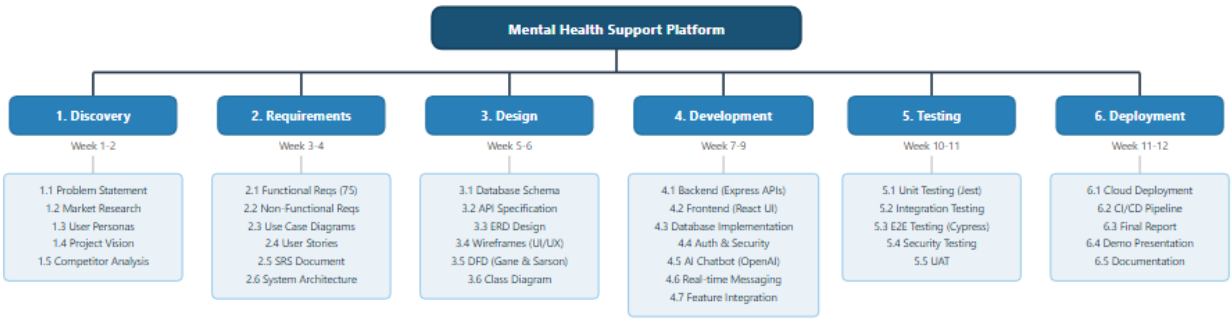

Figure 1.2: Work Breakdown Structure — Mental Health Support Platform

Phase / Timeline	Tasks
1. Discovery (Week 1–2)	Problem Statement, Market Research, User Personas, Project Vision, Competitor Analysis
2. Requirements (Week 3–4)	Functional Requirements (75), Non-Functional Requirements, Use Case Diagrams, User Stories, SRS Document, System Architecture
3. Design (Week 5–6)	Database Schema, API Specification, ERD Design, Wireframes (UI/UX), DFD (Gane & Sarson), Class Diagram
4. Development (Week 7–9)	Backend (Express APIs), Frontend (React UI), Database Implementation, Auth & Security, AI Chatbot (OpenAI), Real-time Messaging, Feature Integration
5. Testing (Week 10–11)	Unit Testing (Jest), Integration Testing, E2E Testing (Cypress), Security Testing, User Acceptance Testing
6. Deployment (Week 11–12)	Cloud Deployment, CI/CD Pipeline, Final Report, Demo Presentation, Documentation

1.11 Gantt Chart

The Gantt chart below maps out the project timeline, showing each phase and the key milestones across our 12-week development period.

Period	Phase	Key Activities	Status
Week 1–2	Discovery & Planning	Problem research, personas, vision	✓
Week 3	Requirements Documentation	SRS, functional/non-functional requirements	✓
Week 4	System Architecture Design	Architecture diagrams, tech stack decisions	✓
Week 5	Database & API Design	ERD, API spec, DFD diagrams	✓
Week 6	UI/UX Design	Wireframes (Balsamiq), design review	✓
Week 7	Backend Development Pt. 1	Auth, user management, appointments APIs	In Progress
Week 8	Backend Development Pt. 2	Mood, messaging, chatbot, emergency APIs	In Progress
Week 9	Frontend Development	React UI, dashboard, all page components	Planned
Week 10	System Integration	Frontend-backend integration, real-time features	Planned

Week 11	Testing & Deployment	Unit, integration, E2E, security testing, deploy	Planned
Week 12	Final Demo & Report	Presentation, documentation, handover	Planned

2. System Requirements

This section lays out what the platform needs to do (functional requirements) and how well it needs to do it (non-functional requirements). We gathered these through the project proposal, discussions with stakeholders, and looking at best practices in HealthTech app development.

2.1 Functional Requirements

In total, we defined 75 functional requirements spread across modules like Authentication, Profile Management, Appointments, Clinical Records, Messaging, Mood Tracking, AI Chatbot, Emergency Support, Notifications, Content Management, Analytics, Administration, System Maintenance, Privacy, and UI/Localization.

FR#	Requirement	CRUD	Module
FR1	Register new user account with unique email/SIM validation	Create	Authentication
FR2	Verify accounts via OTP within a 5-second window	Create	Authentication
FR3	Lock user account after 5 failed login attempts	Update	Authentication
FR4	Record user login timestamp and IP address	Create	Audit
FR5	Process password reset with 10-minute token validity	Update	Authentication
FR6	Update profile fields with strict input validation	Update	Profile
FR7	Upload and store profile image (size < 2MB)	Create	Profile
FR8	User Login/Logout	Read	Authentication
FR9	Schedule appointment with time-slot conflict detection	Create	Appointment
FR10	Reject booking if timeslot is unavailable	Read	Appointment
FR11	Reschedule appointments up to 2 hours before session	Update	Appointment
FR12	Mark appointment as missed after 15-minute delay	Update	Appointment
FR13	Dispatch booking confirmation within 1 minute	Create	Notification
FR14	Approve pending appointments within 10-minute window	Update	Appointment
FR15	Auto-cancel pending bookings after 30 minutes inactivity	Delete	Appointment
FR16	Record clinical session notes post-consultation	Create	Clinical
FR17	Restrict session note access based on consent flags	Read	Security
FR18	View comprehensive patient history (authorized only)	Read	Clinical
FR19	Retain patient records independently of clinician employment	Read	Compliance
FR20	Reassign patients upon clinician departure	Update	Clinical
FR21	Request access to previous medical records	Read	Clinical
FR22	Log all medical record access and viewing actions	Create	Audit
FR23	Encrypt all messages using AES-256	Create	Security
FR24	Deliver chat messages with < 2 seconds latency	Create	Messaging
FR25	Store chat transcripts permanently in database	Create	Database
FR26	Delete chat messages locally (soft delete only)	Delete	Messaging
FR27	Submit daily mood entry with exact timestamp	Create	Mood
FR28	Prevent duplicate mood entries within 1-hour timeframe	Create	Mood
FR29	Generate automated weekly mood reports	Read	Mood
FR30	Trigger risk alert if mood < 2 for 3 consecutive days	Create	AI
FR31	Recommend resources based on mood patterns	Read	AI
FR32	Respond to user queries via Chatbot within 3 seconds	Create	AI Chatbot
FR33	Escalate Chatbot session if confidence < 70%	Update	AI Chatbot
FR34	Record all Chatbot conversation transcripts	Create	AI Chatbot
FR35	Review Chatbot logs via admin dashboard	Read	Admin
FR36	Identify users with 7 days of inactivity	Read	AI
FR37	Dispatch automated wellness check notifications	Create	Notification
FR38	Activate emergency button with < 1 second response	Create	Emergency
FR39	Display crisis hotline numbers on emergency trigger	Read	Emergency
FR40	Log all emergency button activations	Create	Audit
FR41	Add mental health resources via admin panel	Create	Content
FR42	Validate resource URLs before publishing	Create	Content
FR43	Track and aggregate resource view counts	Read	Analytics
FR44	Generate monthly platform analytics reports	Read	Analytics
FR45	Execute automated daily database backups at midnight	Create	System
FR46	Restore database from backup within 1-hour window	Read	System

FR47	Log system errors and exceptions automatically	Create	Monitoring
FR48	Dispatch instant alerts for critical system errors	Create	Monitoring
FR49	Select preferred interface language	Update	UI
FR50	Load main dashboard within 3 seconds	Read	Performance
FR51	Upload clinical documents and worksheets	Create	Clinical
FR52	Validate document uploads (PDF format only)	Create	Clinical
FR53	Download personal medical records securely	Read	Compliance
FR54	Revoke data sharing consent instantaneously	Update	Security
FR55	Restrict all access upon consent revocation	Update	Security
FR56	Log all modifications to user consent settings	Create	Audit
FR57	Integrate external SMS API for notifications	Create	Integration
FR58	Retry failed API calls up to 3 times	Update	Integration
FR59	Queue notifications during service downtimes	Create	Integration
FR60	Deactivate user accounts via admin panel	Update	Admin
FR61	Track active session duration per user	Read	Analytics
FR62	Auto-logout after 15 minutes of inactivity	Update	Security
FR63	Validate inputs to prevent SQL injection	Create	Security
FR64	Generate complete audit trails for DB transactions	Create	Audit
FR65	Assign and enforce RBAC	Update	Security
FR66	Deny and log unauthorized API requests	Read	Security
FR67	Synchronize appointments with Google Calendar	Create	Integration
FR68	Dispatch follow-up for missed appointments	Create	Notification
FR69	Rebook from missed session alerts	Create	Appointment
FR70	Track clinician workload metrics	Read	Analytics
FR71	Generate regulatory compliance reports	Read	Compliance
FR72	Log complete history of data modifications	Create	Audit
FR73	Restrict file uploads exceeding 5MB	Create	System
FR74	Export reports to CSV	Read	Admin
FR75	Maintain system monthly	Read	System

2.2 Non-Functional Requirements

Performance (NFRP01–NFRP04)

ID	Requirement	Category
NFRP01	Page load time ≤ 3 seconds	Performance
NFRP02	API response 95th percentile within 500ms	Performance
NFRP03	Support 50 concurrent users minimum	Scalability
NFRP04	Database queries complete within 200ms	Performance

Security (NFRS01–NFRS06)

ID	Requirement	Category
NFRS01	Password hashing using bcrypt with 10 salt rounds	Security
NFRS02	JWT tokens with 1-hour expiry	Security
NFRS03	HTTPS/TLS 1.2+ for all communications	Security
NFRS04	AES-256 encryption for data at rest	Security
NFRS05	SQL injection and XSS prevention on all inputs	Security
NFRS06	Rate limiting: 100 requests/IP/minute (HTTP 429)	Security

Usability (NFRU01–NFRU04)

ID	Requirement	Category
NFRU01	Intuitive interface with ≤ 5 minute onboarding	Usability
NFRU02	Responsive design: 320px – 1920px viewport	Usability
NFRU03	WCAG 2.1 Level AA compliance	Accessibility
NFRU04	Descriptive and actionable error messages	Usability

Reliability (NFRR01–NFRR03)

ID	Requirement	Category
NFRR01	99% system uptime	Reliability
NFRR02	Graceful degradation for non-critical features	Reliability
NFRR03	Daily automated backups with 7-day retention	Reliability

Maintainability (NFRM01–NFRM03)

ID	Requirement	Category
NFRM01	ESLint enforced code standards	Maintainability
NFRM02	Modular architecture with clear separation of concerns	Maintainability
NFRM03	Scalable design supporting 10x user growth	Scalability

Privacy & Compliance (NFRC01–NFRC04)

ID	Requirement	Category
NFRC01	Privacy by design principles throughout	Privacy
NFRC02	Granular consent management	Compliance
NFRC03	Data anonymisation for analytics	Privacy
NFRC04	Right to deletion within 30 days	Compliance

3. System Architecture

3.1 Software and Hardware Architecture

Our platform uses a three-tier architecture: a Presentation Tier (the front end), an Application Tier (the server), and a Data Tier (the database). This setup keeps things separated, makes the system easier to scale, and simpler to maintain.

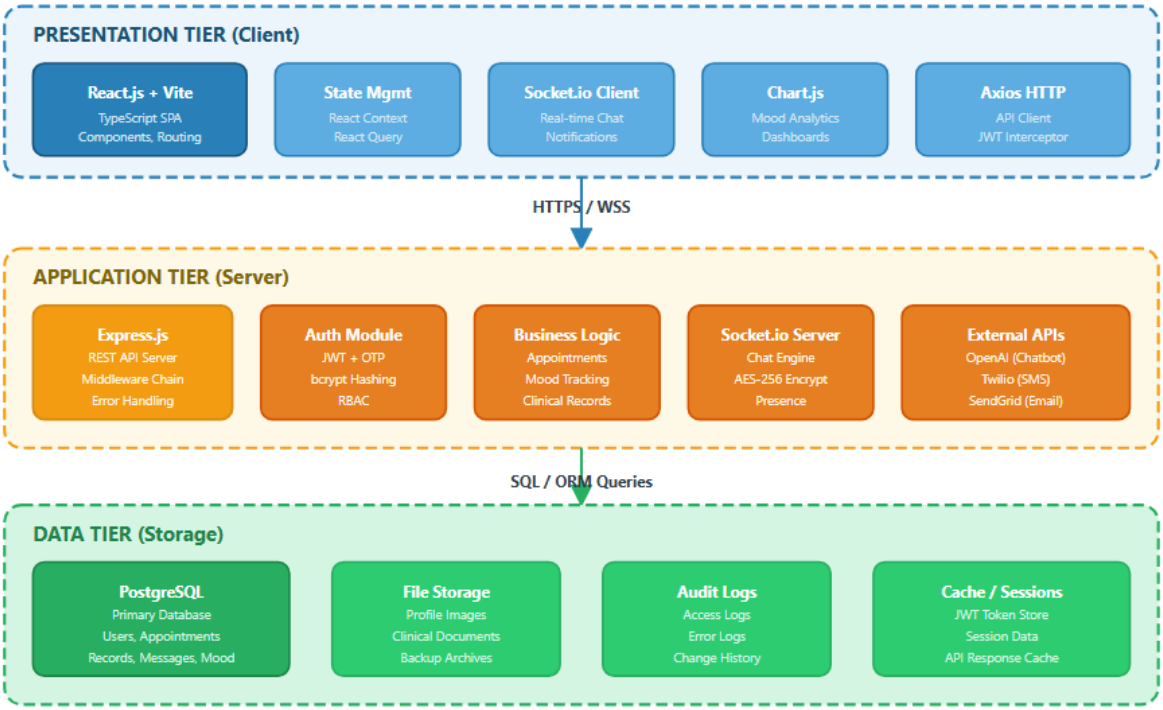

Figure 3.1: Three-Tier System Architecture Diagram

Architecture Explanation:

Presentation Tier (Client): This is the front end, built with React.js and Vite using TypeScript. It is a Single Page Application with component-based routing. We use React Context and React Query to manage state, Socket.io for real-time chat and notifications, Chart.js for mood analytics charts, and Axios with a JWT interceptor for secure API calls.

Application Tier (Server): The back end runs on Express.js with Node.js. It has a REST API with middleware and error handling, an auth module that handles JWT tokens, OTP, bcrypt hashing, and role-based access. The business logic covers appointments, mood tracking, and clinical records. We also run a Socket.io server for real-time chat with AES-256 encryption, and connect to external services like OpenAI for the chatbot, Twilio for SMS, and SendGrid for email.

Data Tier (Storage): PostgreSQL is our main database, storing users, appointments, clinical records, messages, and mood entries. We also use file storage for profile pictures, clinical documents, and backup

files. Audit logs track all access, errors, and changes, while a caching layer handles JWT tokens, session data, and API response caching.

Communication: The front end talks to the server over HTTPS and WSS (WebSocket Secure). The server communicates with the database through SQL queries via an ORM.

3.2 Tech Stack (Detailed)

Component	Technology	Purpose
Frontend Framework	React.js 18+ with Vite	Component-based SPA with TypeScript support
State Management	React Context + React Query	Server state caching and local state management
UI Styling	Tailwind CSS / Material UI	Responsive design, utility-first CSS
Charts & Visualization	Chart.js / Recharts	Mood trend visualization, analytics dashboards
Real-time Communication	Socket.io (Client)	WebSocket-based chat and notifications
HTTP Client	Axios	API communication with JWT interceptor
Backend Runtime	Node.js 18+	Server-side JavaScript runtime
API Framework	Express.js	RESTful API with middleware architecture
Authentication	JWT + bcrypt + OTP	Stateless auth, password hashing, 2FA
Chat Encryption	AES-256	End-to-end message encryption
Database	PostgreSQL 15+	Primary relational database (ACID compliant)
ORM	Prisma / Sequelize	Database abstraction and migration management
AI/NLP	OpenAI API (GPT-3.5/4)	Chatbot NLP processing
SMS Gateway	Twilio	SMS notifications and OTP delivery
Email Service	SendGrid	Transactional email notifications
Calendar Integration	Google Calendar API	Appointment synchronization
Testing (Unit)	Jest	Unit test framework for Node.js and React
Testing (E2E)	Cypress	End-to-end browser testing
Version Control	Git / GitHub	Source code management and collaboration
CI/CD	GitHub Actions	Automated testing and deployment pipeline
Cloud Hosting	Render / AWS	Application and database hosting
Wireframing	Balsamiq Cloud	Low-fidelity wireframe design

4. System Design

4.1 User Interface Design / Wireframes

We designed the wireframes below in Balsamiq Cloud. They show the main screens of the platform, including page layouts, navigation, and the key interactive elements.

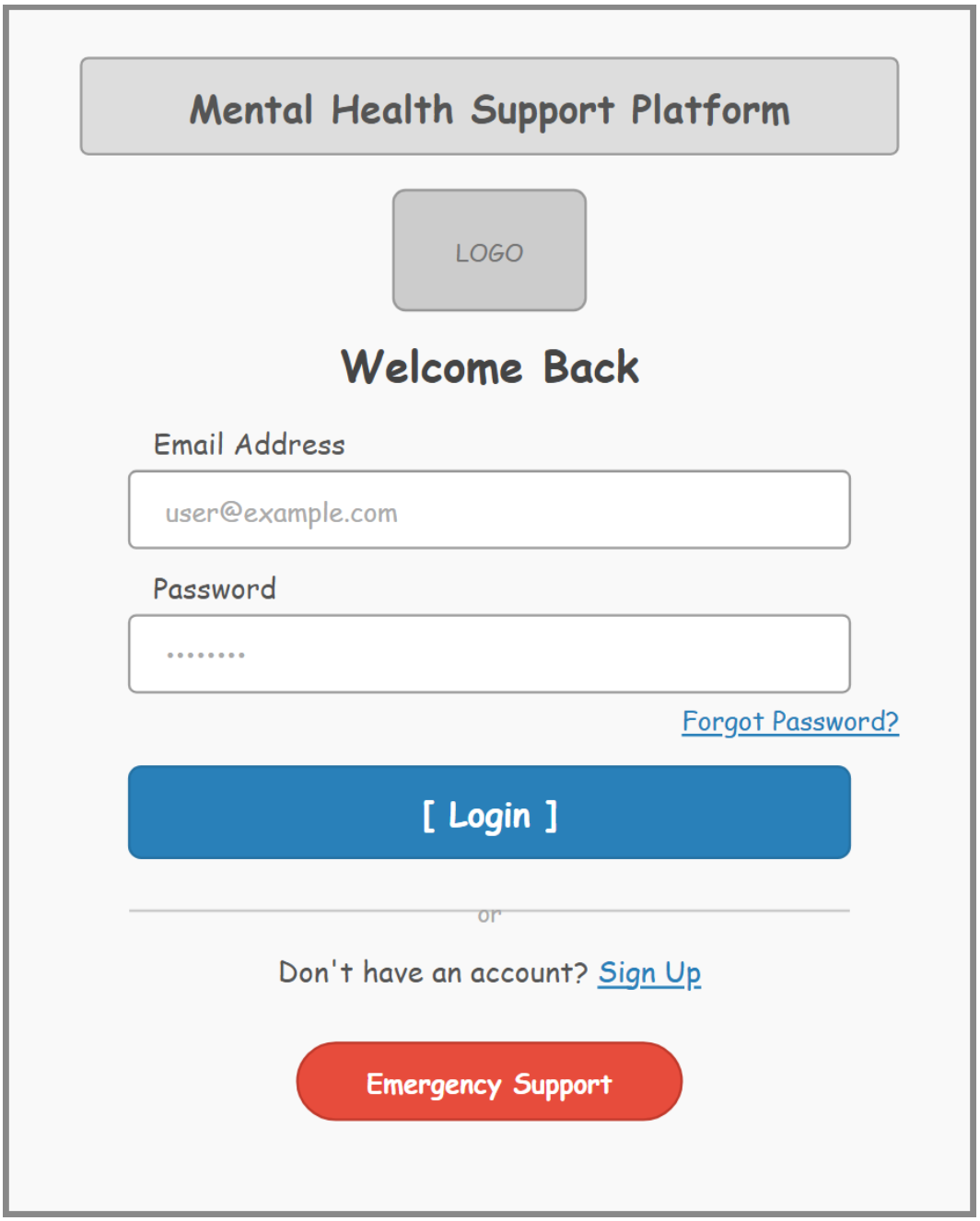

Figure 4.1: Login / Registration Page

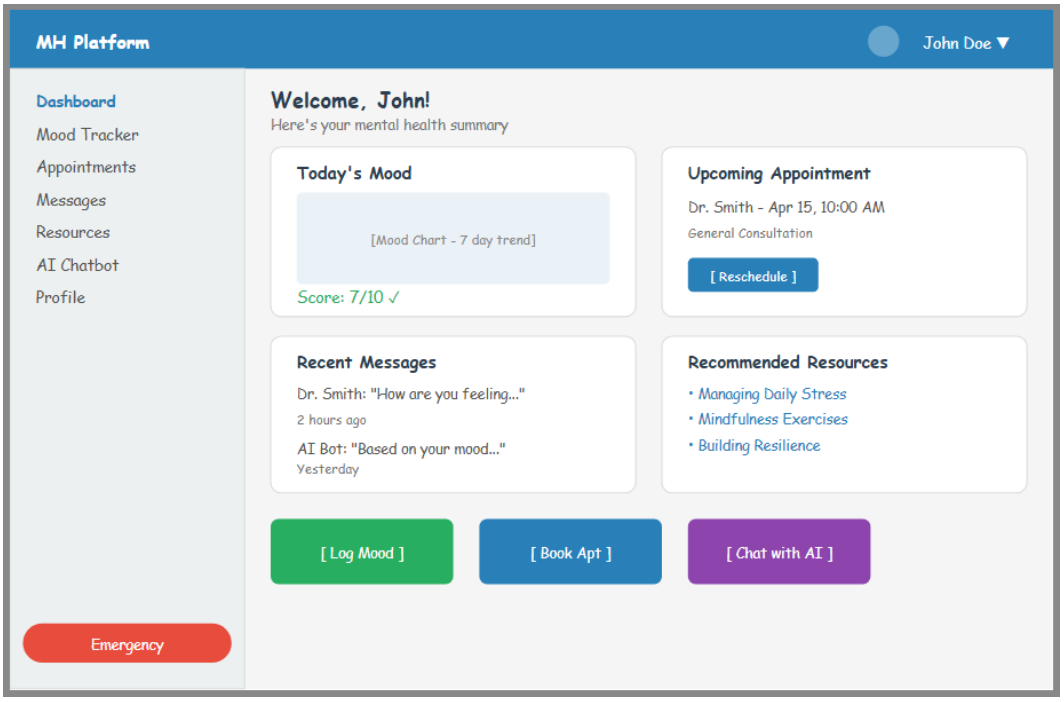

Figure 4.2: User Dashboard

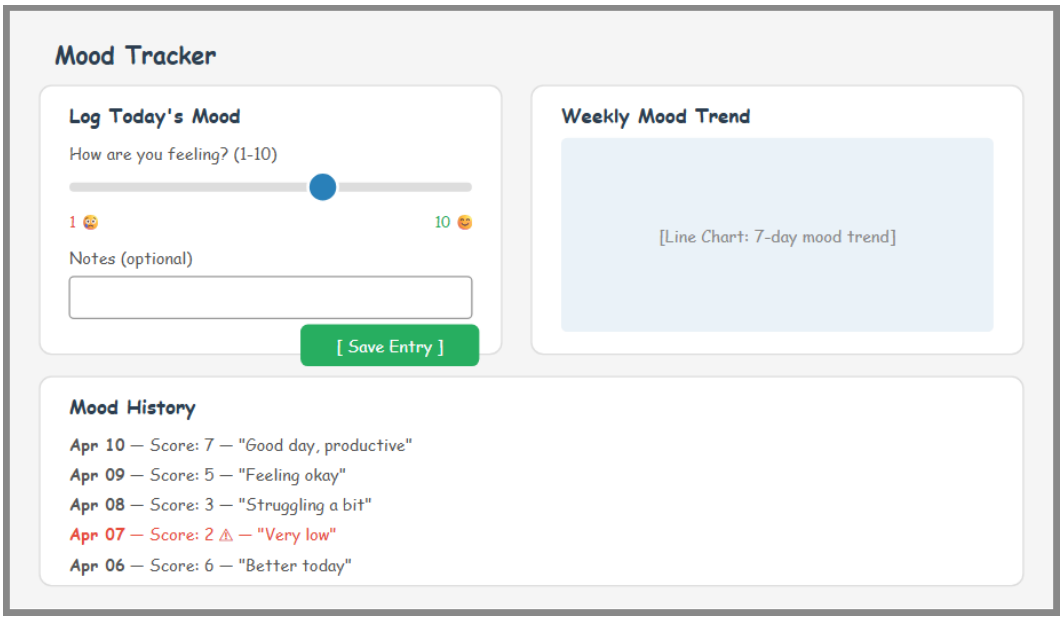

Figure 4.3: Mood Tracker Page

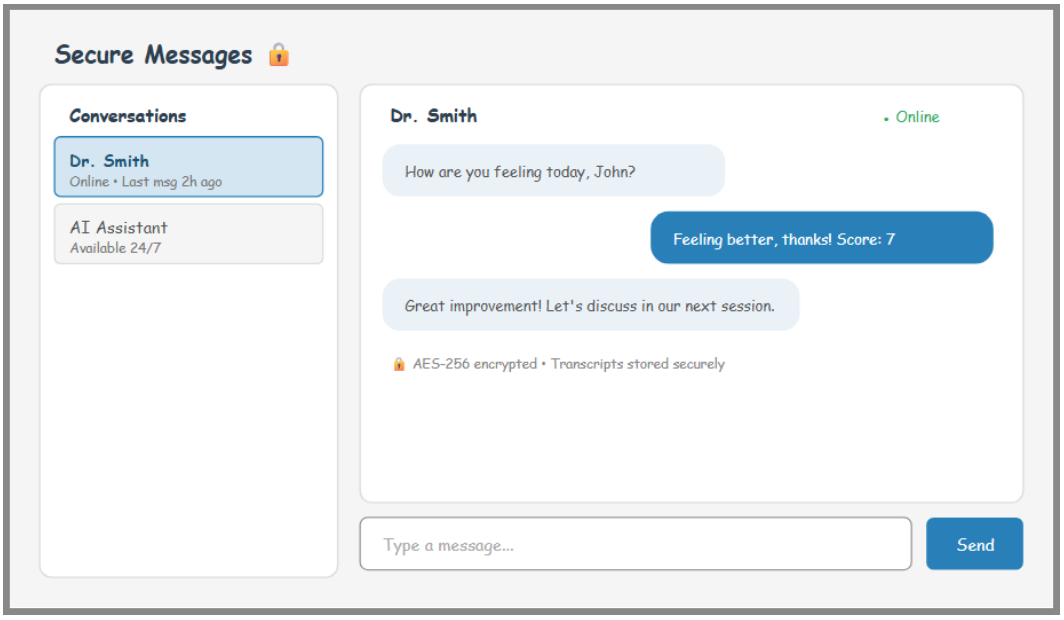

Figure 4.4: Secure Messaging Interface

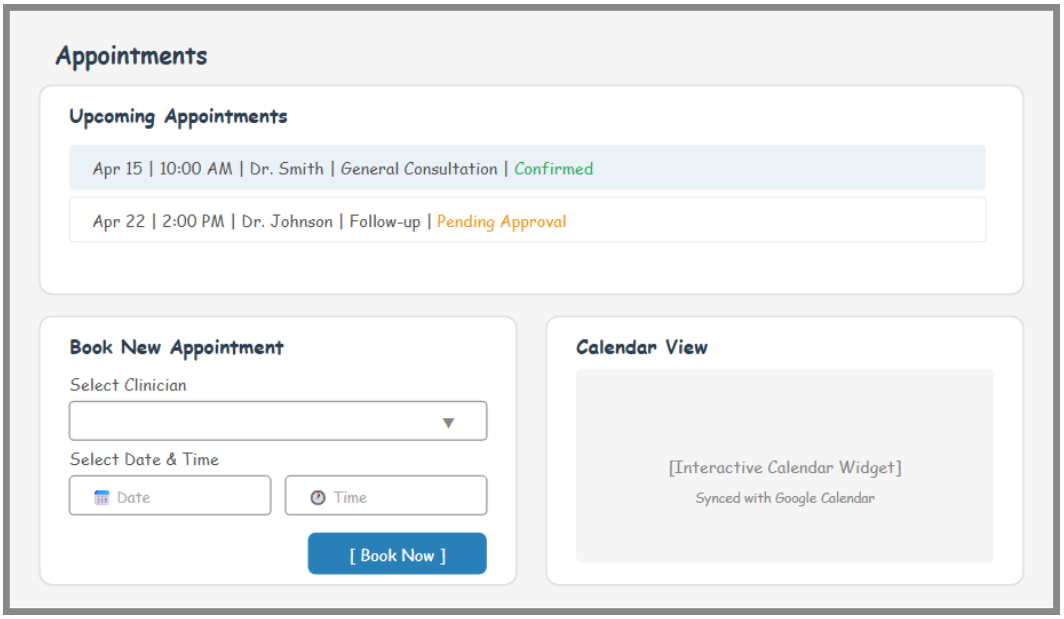

Figure 4.5: Appointments Page

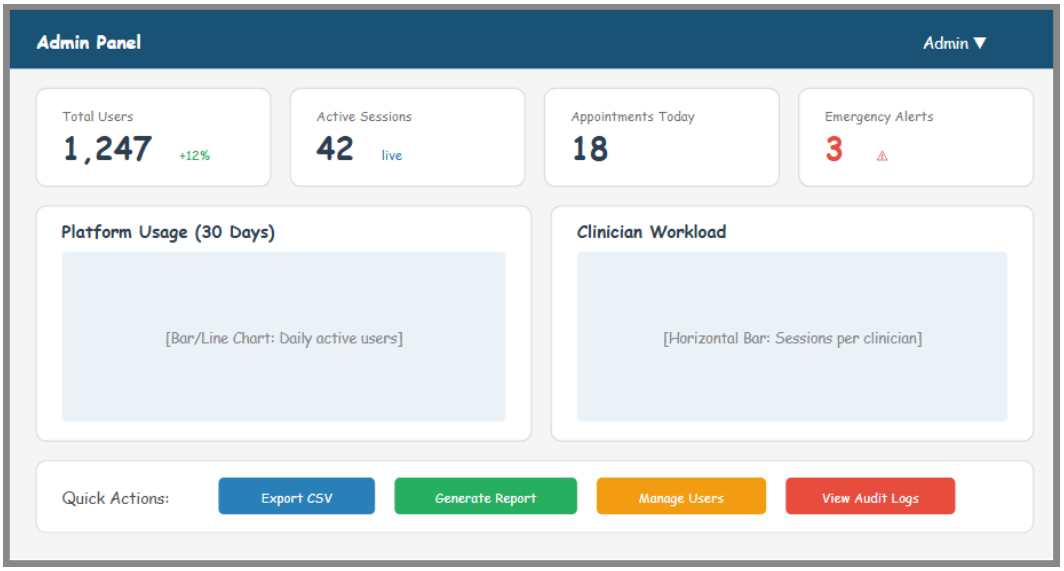

Figure 4.6: Admin Dashboard Panel

5. Use Case Diagrams

The use case diagrams below show how the three main actors — Patient/User, Clinician/Doctor, and Administrator — interact with the platform. Each use case links back to a specific functional requirement.

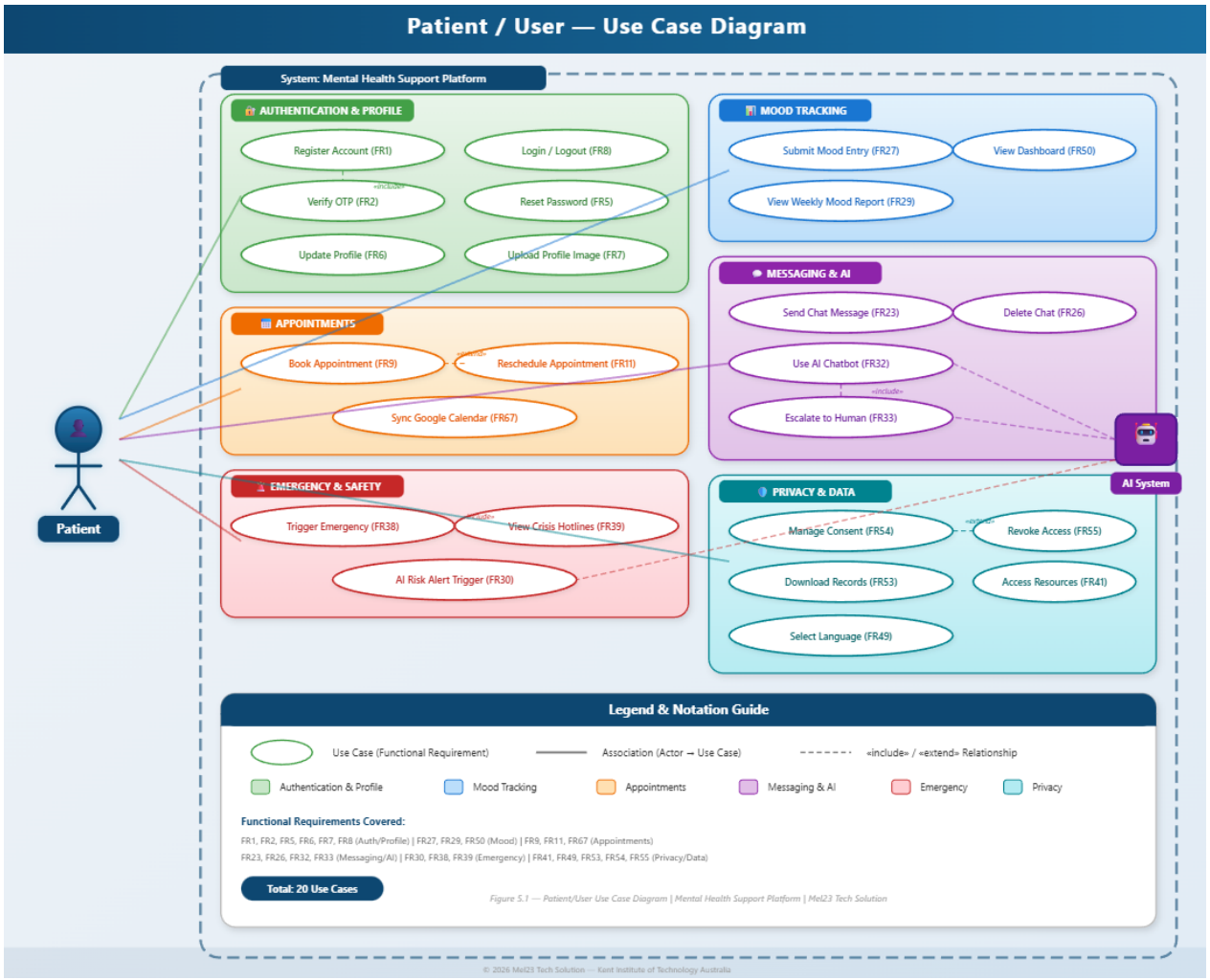

Figure 5.1: Patient / User — Use Case Diagram

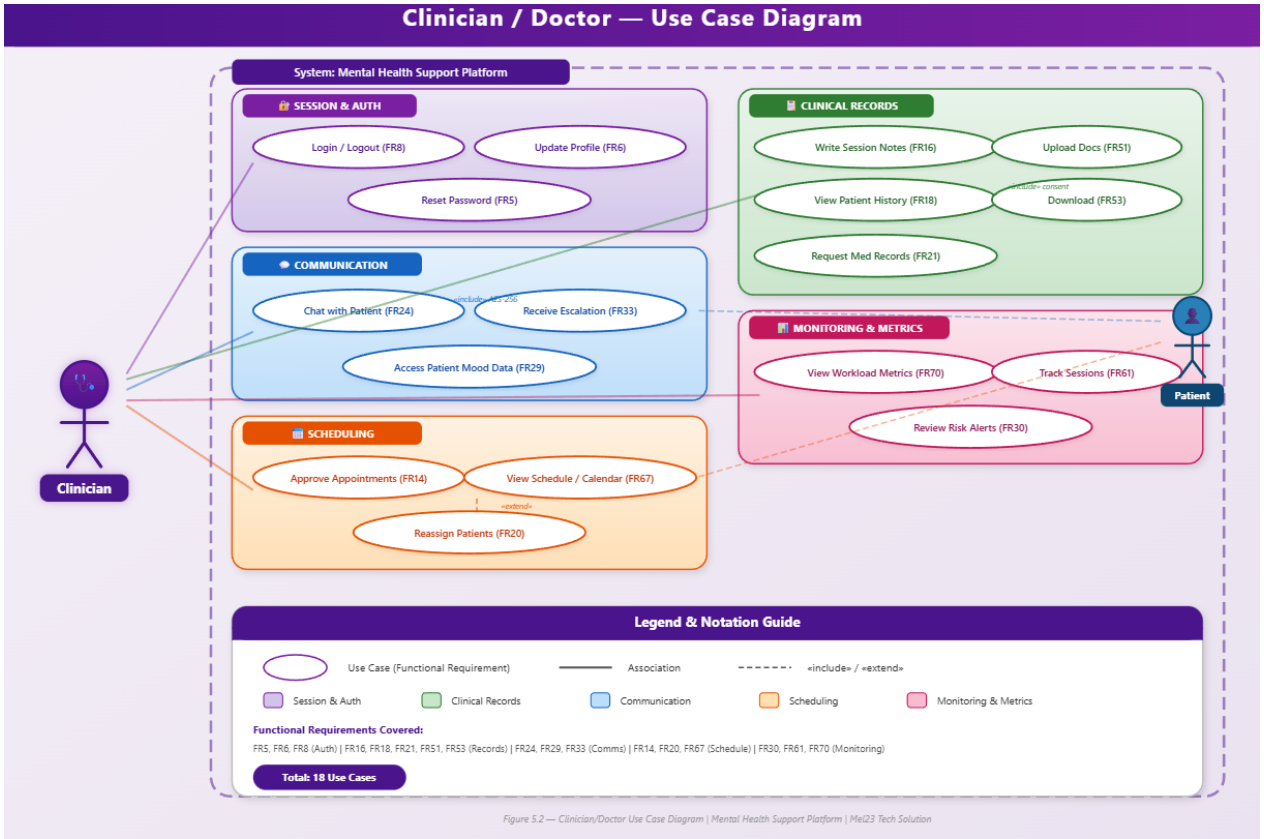

Figure 5.2: Clinician / Doctor — Use Case Diagram

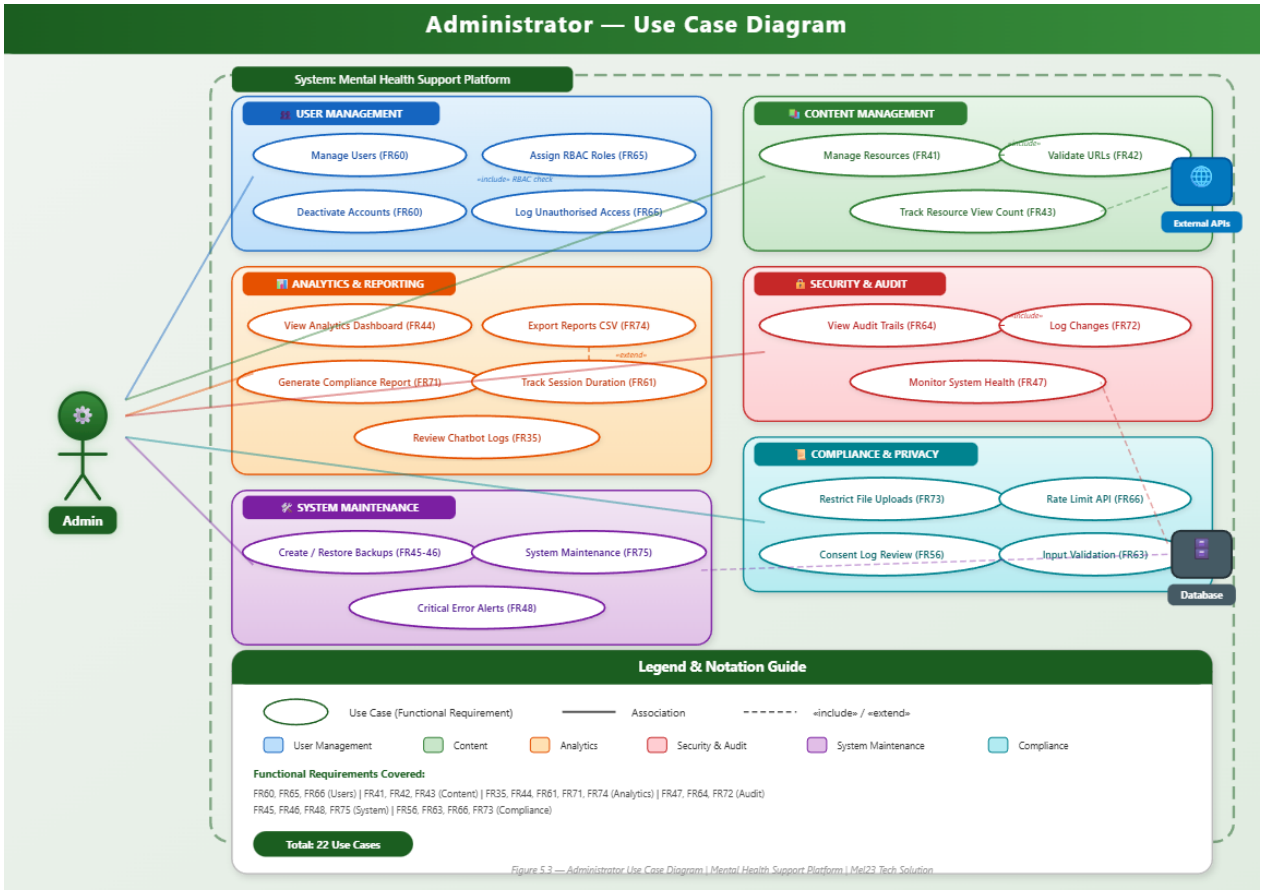

Figure 5.3: Administrator — Use Case Diagram

Use Case Summary — Patient/User:

- Register Account (FR1), Login/Logout (FR8), Verify OTP (FR2), Reset Password (FR5)
- Update Profile (FR6), Upload Profile Image (FR7)
- Submit Mood Entry (FR27), View Mood Report (FR29)
- Book Appointment (FR9), Reschedule Appointment (FR11)
- Send Chat Message (FR23), Use AI Chatbot (FR32)
- Access Resources (FR41), Trigger Emergency (FR38)
- Manage Consent (FR54), View Dashboard (FR50), Download Records (FR53), Select Language (FR49)

Use Case Summary — Clinician:

- Login/Logout (FR8), View Patient History (FR18), Write Session Notes (FR16)
- Approve Appointments (FR14), Chat with Patient (FR24), View Schedule (FR67)
- Request Medical Records (FR21), Reassign Patients (FR20)
- Upload Clinical Docs (FR51), View Workload Metrics (FR70), Access Patient Mood Data (FR29)

Use Case Summary — Administrator:

- Manage Users (FR60), Assign RBAC Roles (FR65), Manage Resources (FR41), Validate URLs (FR42)

- View Analytics Dashboard (FR44), Generate Compliance Reports (FR71)
- Review Chatbot Logs (FR35), Monitor System Health (FR47)
- Create/Restore Backups (FR45–46), Export Reports to CSV (FR74)
- Deactivate Accounts (FR60), View Audit Trails (FR64)

6. Context Diagrams (Gane & Sarson Method)

The Level 0 Context Diagram gives a bird's-eye view of the whole system and how it connects to outside entities. We used the Gane & Sarson style, where processes are shown as rounded rectangles and external entities are plain rectangles.

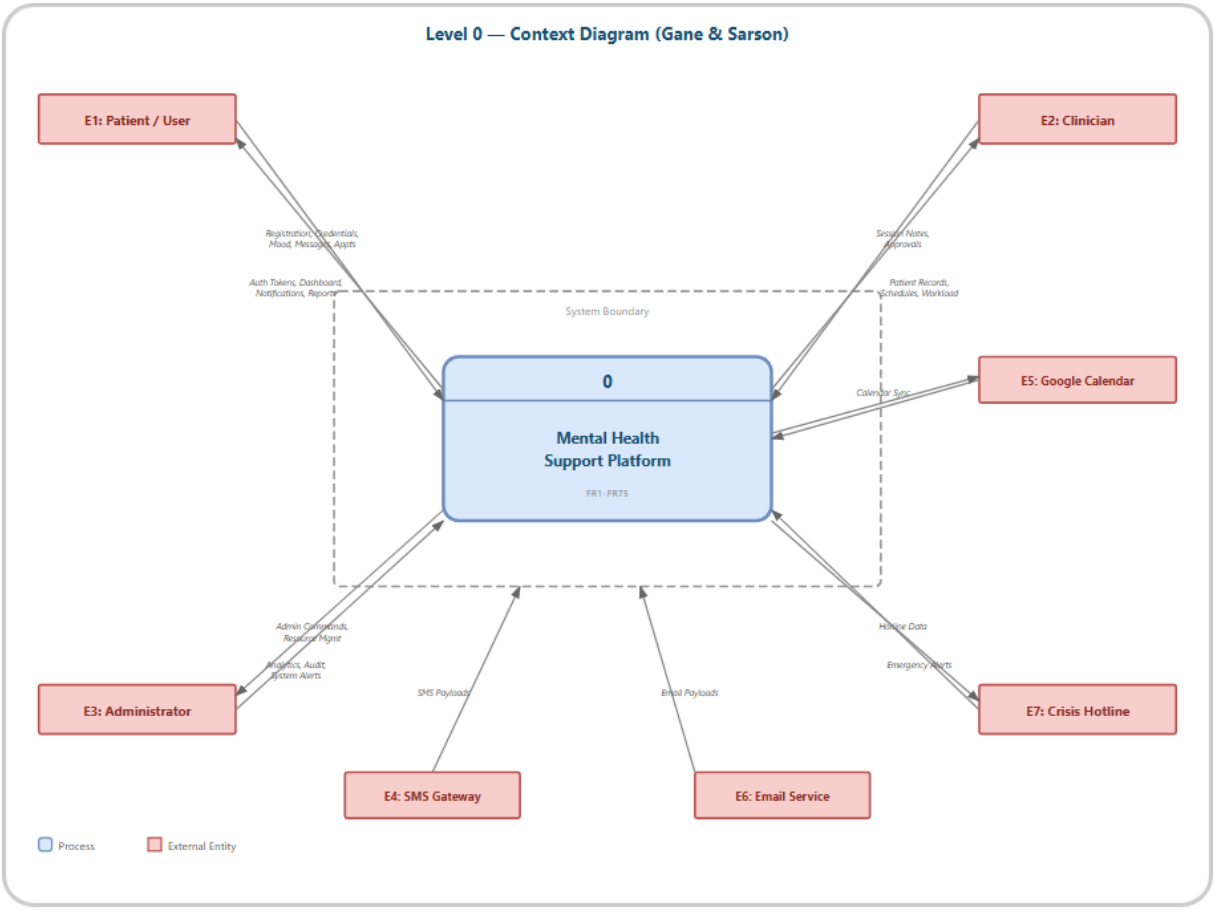

Figure 6.1: Level 0 — Context Diagram (Gane & Sarson)

External Entities:

- E1: Patient/User — The main user who signs up, logs mood, books appointments, and browses resources
- E2: Clinician — A health professional who handles appointments, writes session notes, and checks patient records
- E3: Administrator — The system admin who manages users, resources, analytics, and settings
- E4: SMS Gateway (Twilio) — An outside service we use for sending text messages and OTP codes
- E5: Google Calendar — Used to sync appointments with the user's calendar
- E6: Email Service (SendGrid) — Handles outgoing emails like booking confirmations and alerts
- E7: Crisis Hotline — An external service that provides emergency hotline information

7. Data Flow Diagrams (Gane & Sarson Method)

The Data Flow Diagrams (DFDs) break the system down level by level, showing how data moves between processes, outside entities, and data stores. Everything follows the Gane & Sarson notation.

7.1 DFD Level 1

The Level 1 DFD takes the single system process and splits it into 15 sub-processes, each tied to specific functional requirements. It maps the data flows between external entities (E1–E7), processes (1.0–15.0), and 15 data stores (D1–D15).

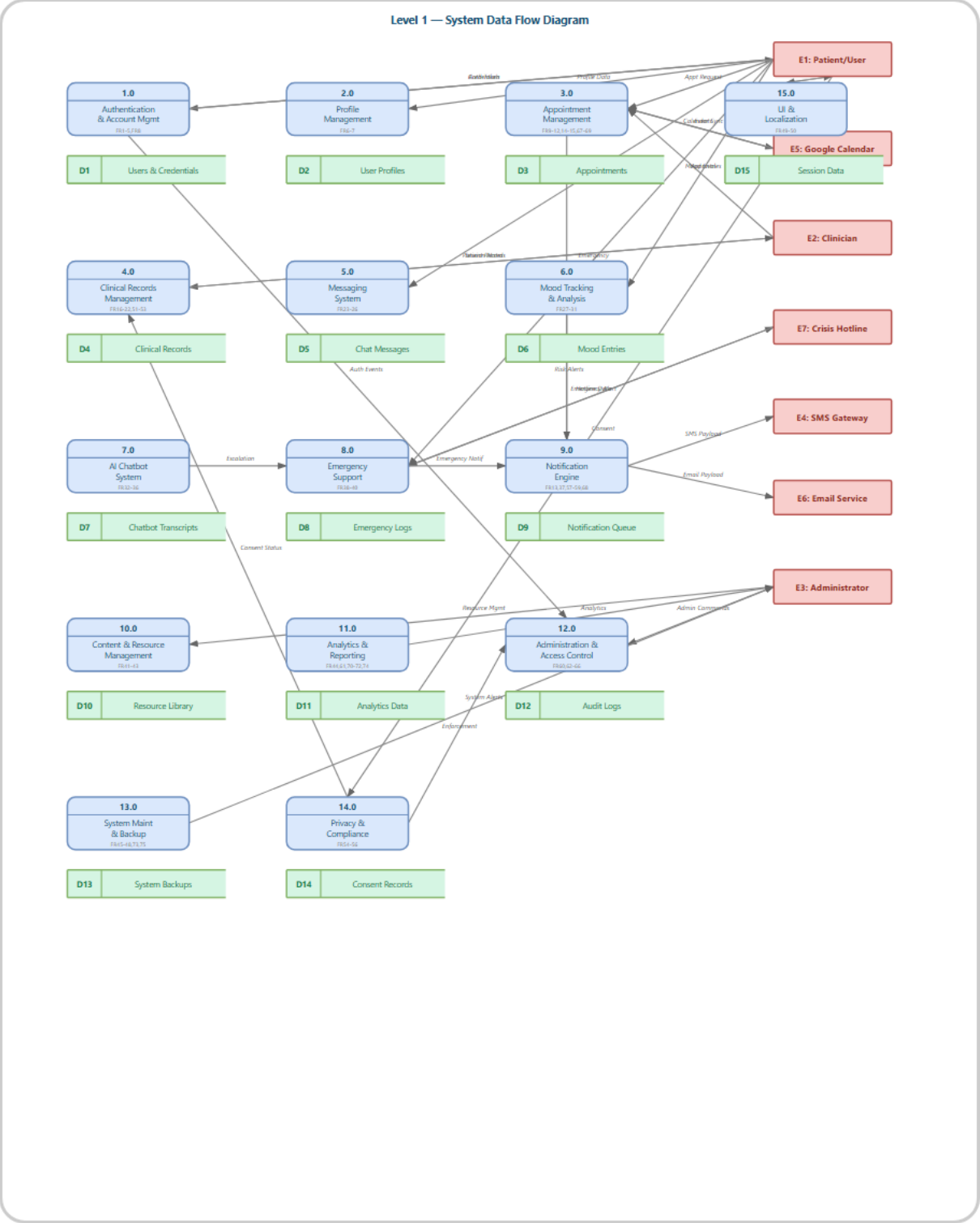

Figure 7.1: Level 1 — System Data Flow Diagram

Process	Name	FRs Covered
1.0	Authentication & Account Management	FR1–5, FR8
2.0	Profile Management	FR6–7
3.0	Appointment Management	FR9–15, FR67–69

4.0	Clinical Records Management	FR16–22, FR51–53
5.0	Messaging System	FR23–26
6.0	Mood Tracking & Analysis	FR27–31
7.0	AI Chatbot System	FR32–36
8.0	Emergency Support	FR38–40
9.0	Notification Engine	FR13, FR37, FR57–59, FR68
10.0	Content & Resource Management	FR41–43
11.0	Analytics & Reporting	FR44, FR61, FR70–72, FR74
12.0	Administration & Access Control	FR60, FR62–66
13.0	System Maintenance & Backup	FR45–48, FR73, FR75
14.0	Privacy & Compliance	FR54–56
15.0	UI & Localization	FR49–50

Data Stores:

Store ID	Name
D1	Users & Credentials
D2	User Profiles
D3	Appointments
D4	Clinical Records
D5	Chat Messages
D6	Mood Entries
D7	Chatbot Transcripts
D8	Emergency Logs
D9	Notification Queue
D10	Resource Library
D11	Analytics Data
D12	Audit Logs
D13	System Backups
D14	Consent Records
D15	Session Data

7.2 DFD Level 2

The Level 2 DFDs drill deeper into each Level 1 process. They show the internal data flows, decision points, and which data stores are used within each process.

Breaks down Process 1.0 into 7 steps: registering a user (FR1), verifying the OTP (FR2), locking accounts (FR3), logging login activity (FR4), resetting passwords (FR5), login/logout (FR8), and deleting and anonymising data (FR8d).

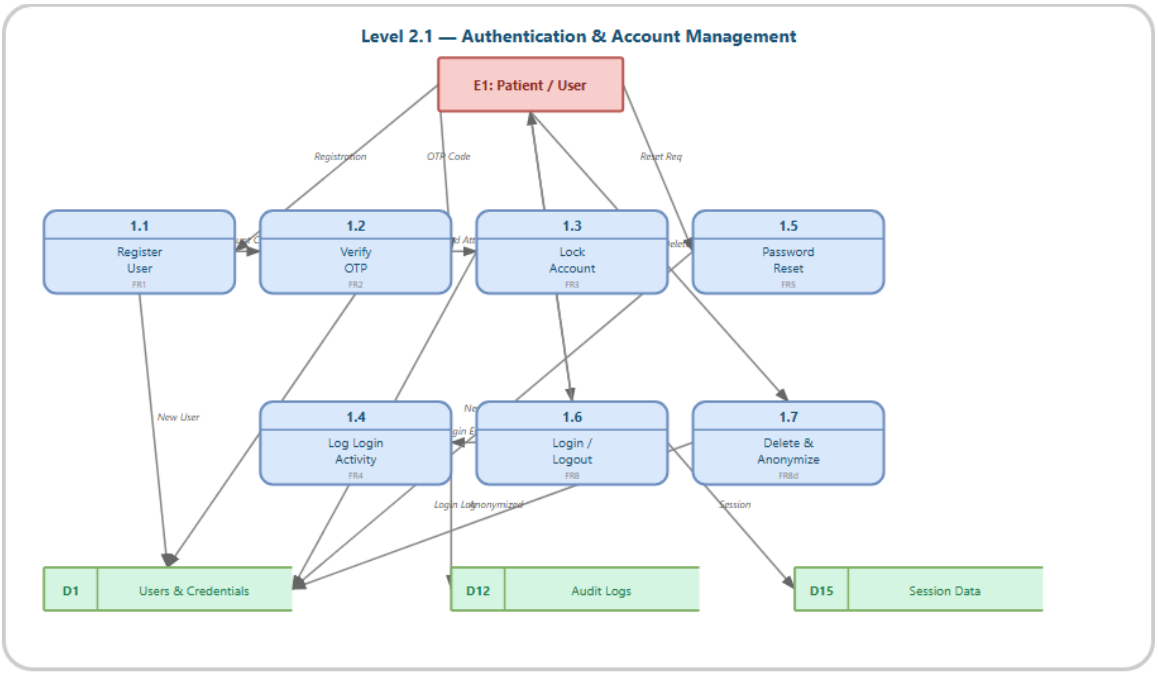

Figure 7.2.1: Level 2.1 — Authentication & Account Management

Breaks down Process 2.0 into 2 steps: updating profile details (FR6) and uploading a profile image with a file-size check under 2 MB (FR7).

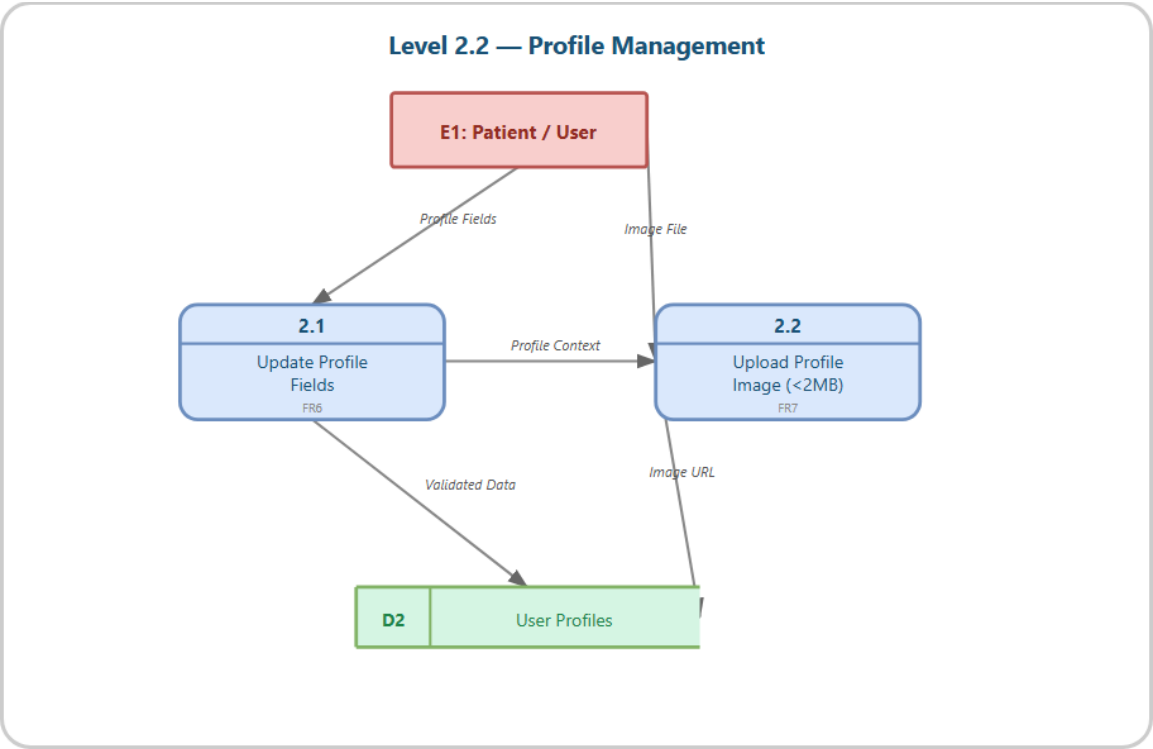

Figure 7.2.2: Level 2.2 — Profile Management

Breaks down Process 3.0 into 9 steps covering things like scheduling (FR9), checking slot availability (FR10), rescheduling (FR11), marking missed appointments (FR12), pending approvals (FR14), auto-cancel (FR15), Google Calendar sync (FR67), follow-up notifications (FR68), and rebooking from alerts (FR69).

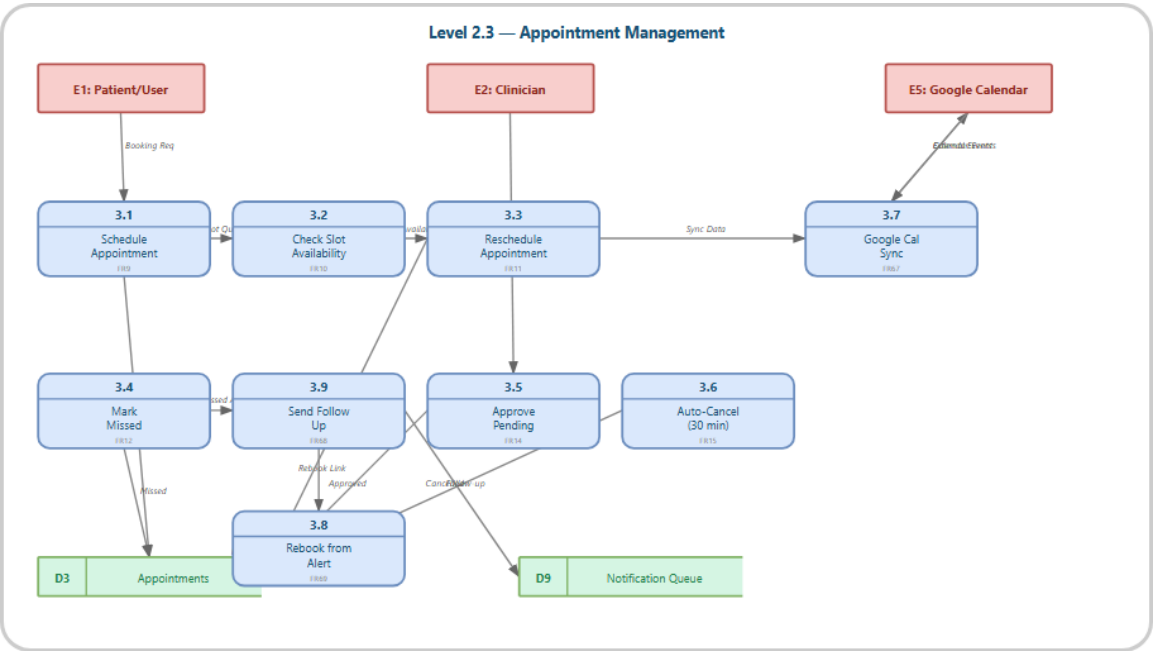

Figure 7.2.3: Level 2.3 — Appointment Management

Breaks down Process 4.0 into 10 steps: writing session notes (FR16), checking consent (FR17), viewing patient history (FR18), managing record retention (FR19), reassigning patients (FR20), handling medical record requests (FR21), logging access (FR22), uploading clinical documents (FR51), validating file formats (FR52), and downloading records (FR53).

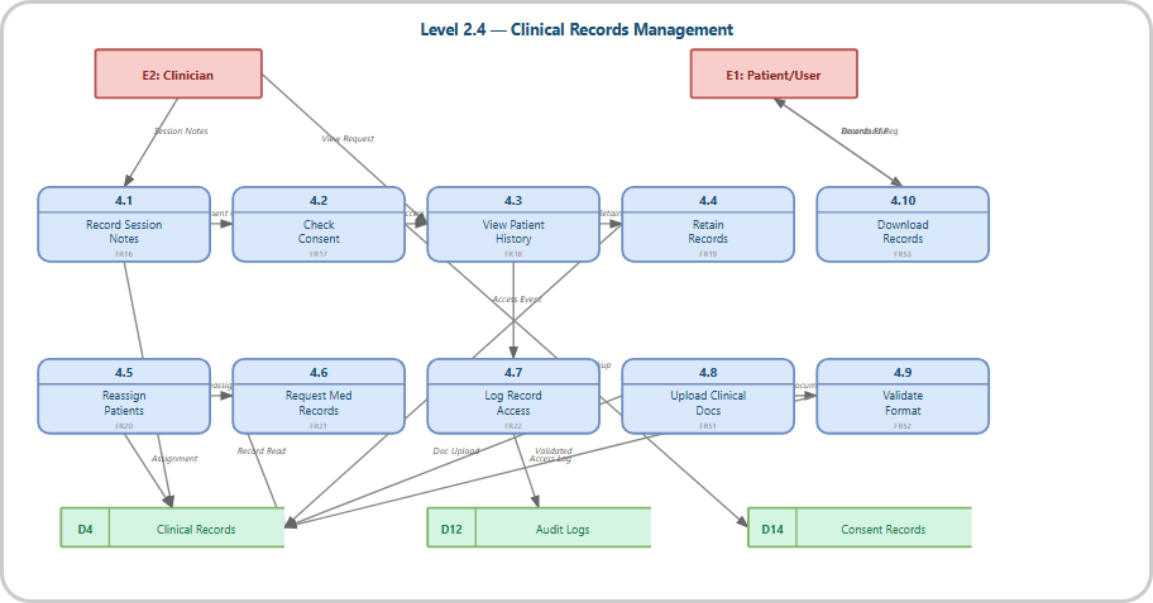

Figure 7.2.4: Level 2.4 — Clinical Records Management

Breaks down Process 5.0 into 4 steps: encrypting messages with AES-256 (FR23), delivering them within 2 seconds (FR24), storing chat transcripts (FR25), and soft deleting messages (FR26).

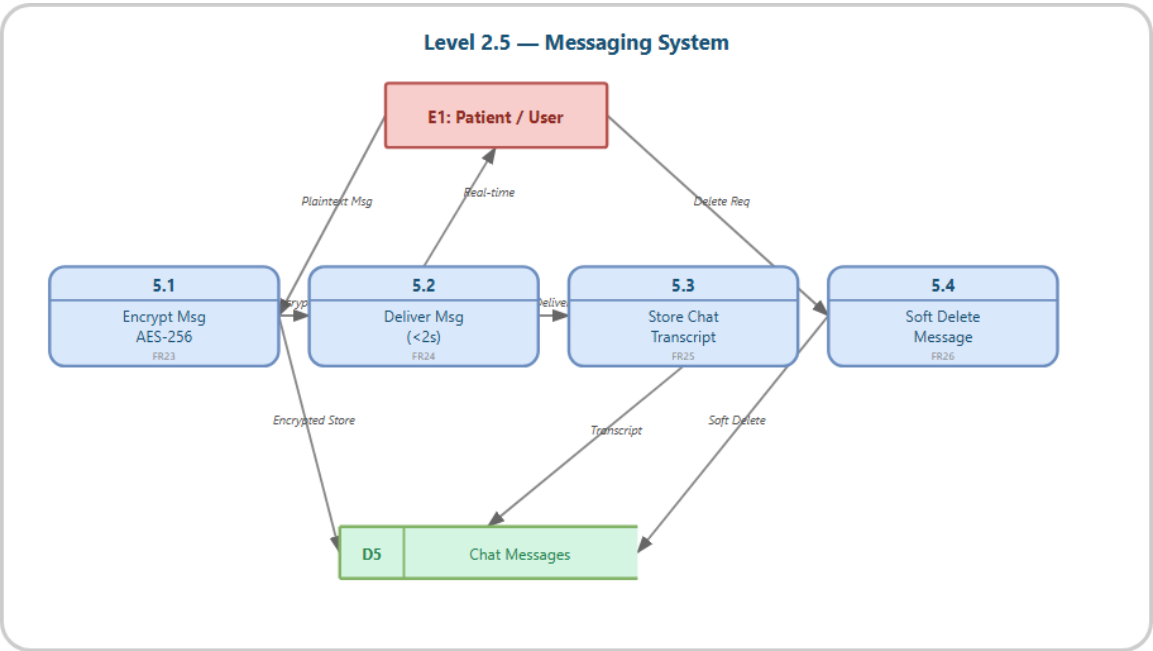

Figure 7.2.5: Level 2.5 — Messaging System

Breaks down Process 6.0 into 5 steps: submitting a mood entry (FR27), preventing duplicates within a 1-hour window (FR28), generating weekly mood reports (FR29), triggering risk alerts when scores are low several days in a row (FR30), and recommending resources based on mood patterns (FR31).

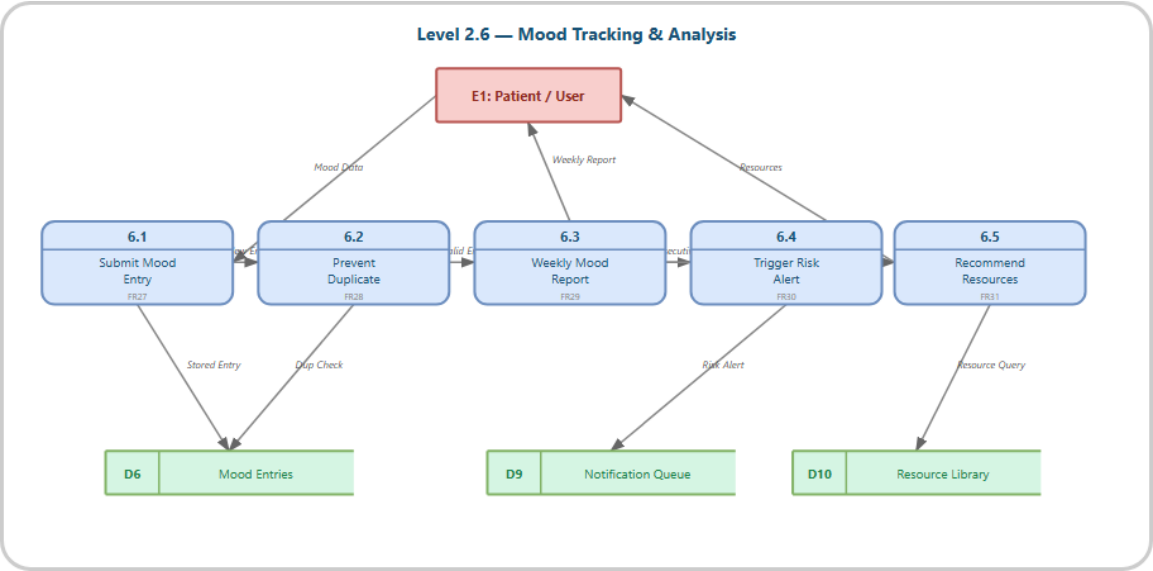

Figure 7.2.6: Level 2.6 — Mood Tracking & Analysis

Breaks down Process 7.0 into 5 steps: processing a query within 3 seconds (FR32), escalating to a human when confidence is below 70% (FR33), saving transcripts (FR34), letting admins review logs (FR35), and flagging users who have been inactive for 7 days (FR36).

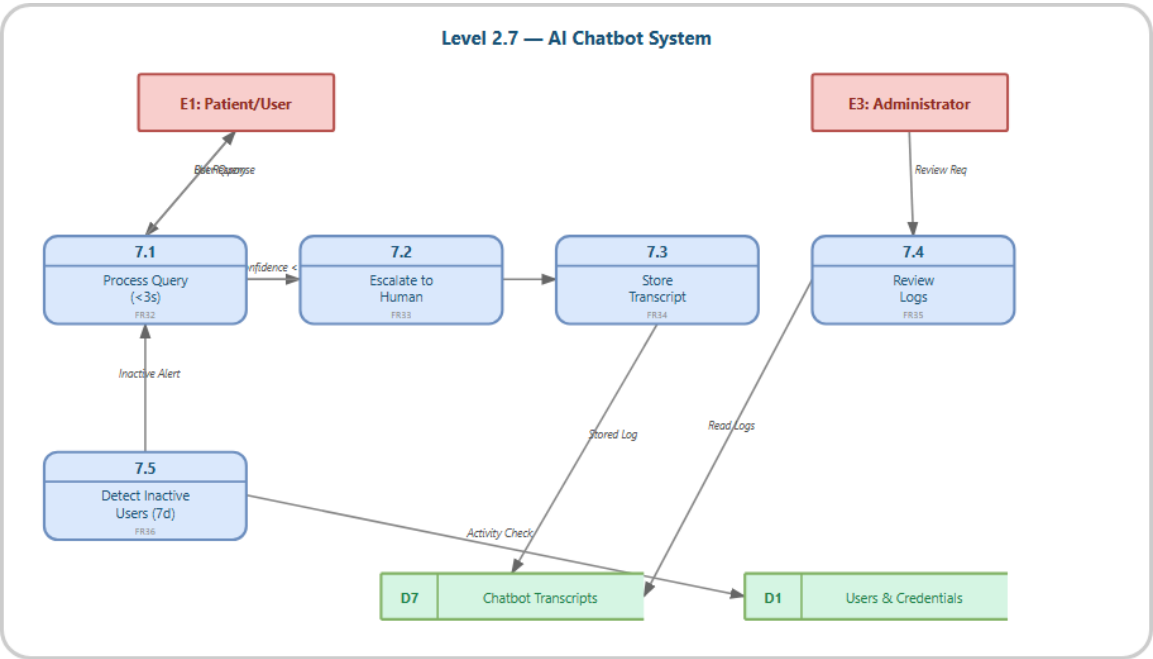

Figure 7.2.7: Level 2.7 — AI Chatbot System

Breaks down Process 8.0 into 3 steps: activating the emergency button in under 1 second (FR38), showing crisis hotline info (FR39), and logging the activation (FR40).

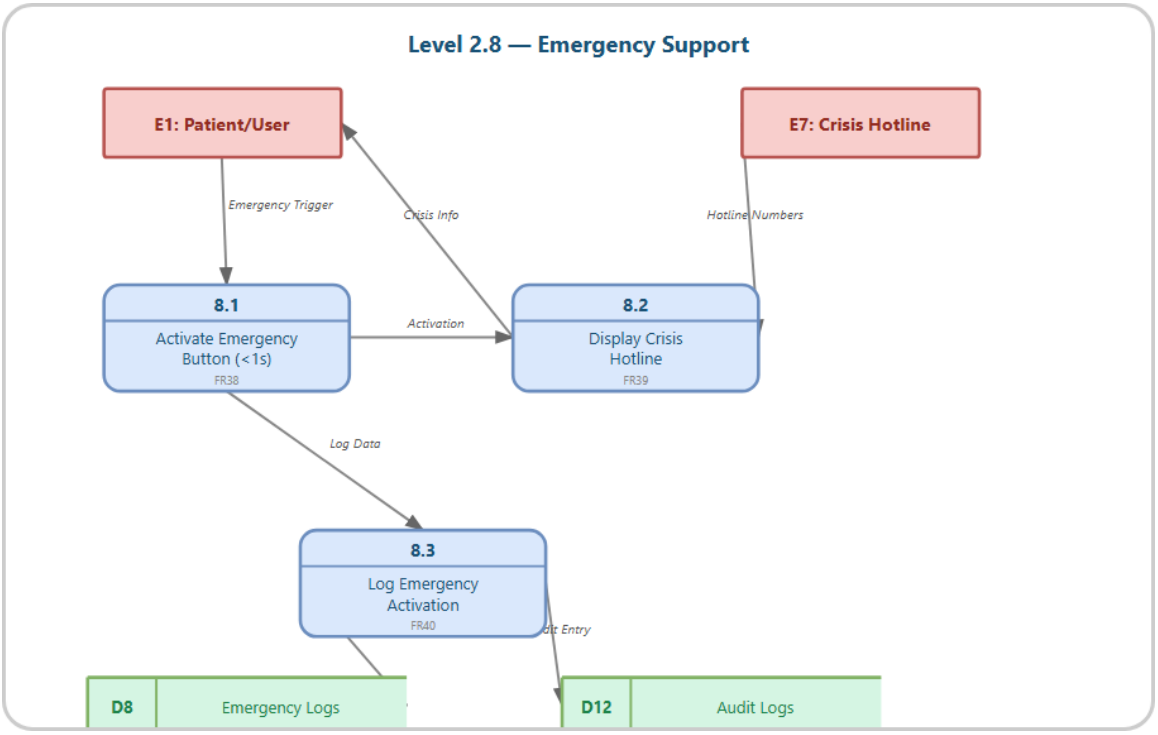

Figure 7.2.8: Level 2.8 — Emergency Support

Breaks down Process 9.0 into 6 steps: sending a booking confirmation (FR13), dispatching wellness checks (FR37), integrating SMS and email (FR57), retrying failed sends up to 3 times (FR58), queuing messages during downtime (FR59), and following up on missed appointments (FR68).

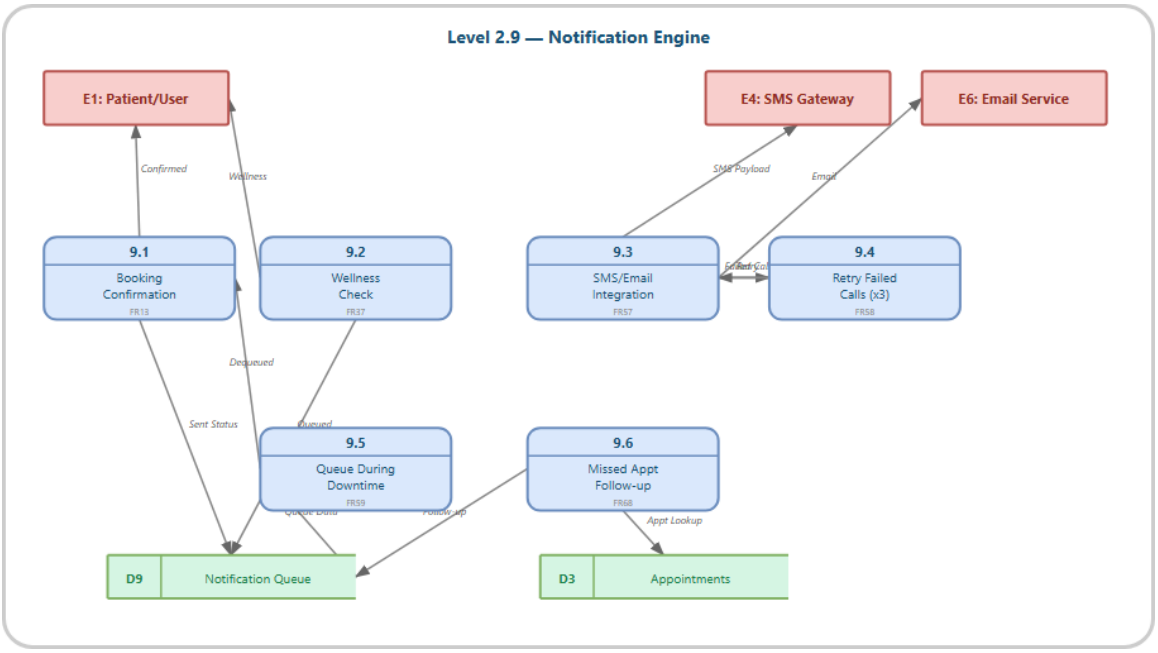

Figure 7.2.9: Level 2.9 — Notification Engine

Breaks down Process 10.0 into 3 steps: adding resources through the admin panel (FR41), checking URLs before they go live (FR42), and tracking how many times each resource is viewed (FR43).

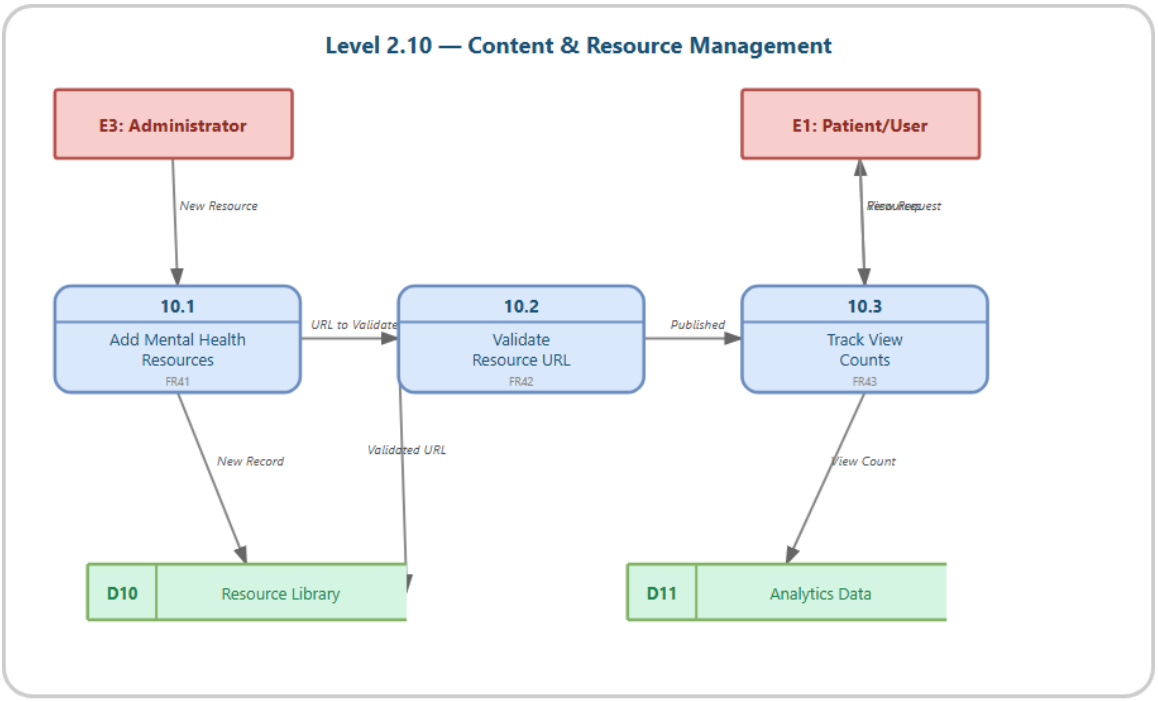

Figure 7.2.10: Level 2.10 — Content & Resource Management

Breaks down Process 11.0 into 6 steps: generating monthly reports (FR44), tracking session times (FR61), evaluating clinician workloads (FR70), producing compliance reports (FR71), logging data changes (FR72), and exporting to CSV (FR74).

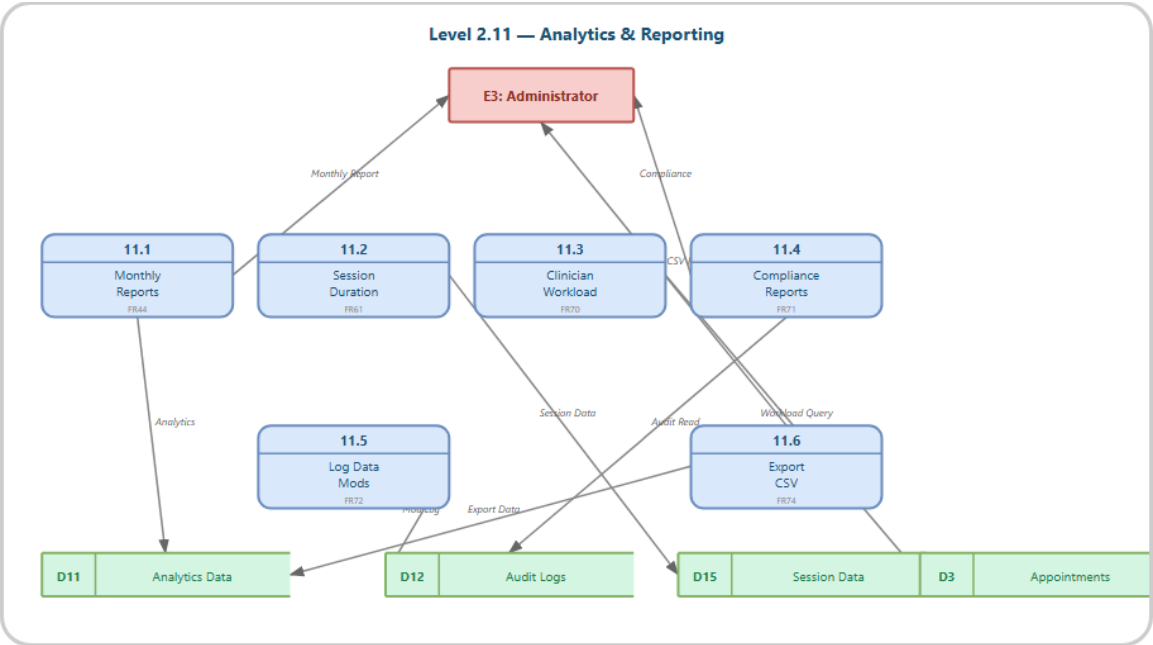

Figure 7.2.11: Level 2.11 — Analytics & Reporting

Breaks down Process 12.0 into 6 steps: deactivating accounts (FR60), auto-logging out after 15 minutes of inactivity (FR62), validating inputs to stop SQL injection (FR63), creating audit trails (FR64), assigning roles via RBAC (FR65), and blocking unauthorised API requests (FR66).

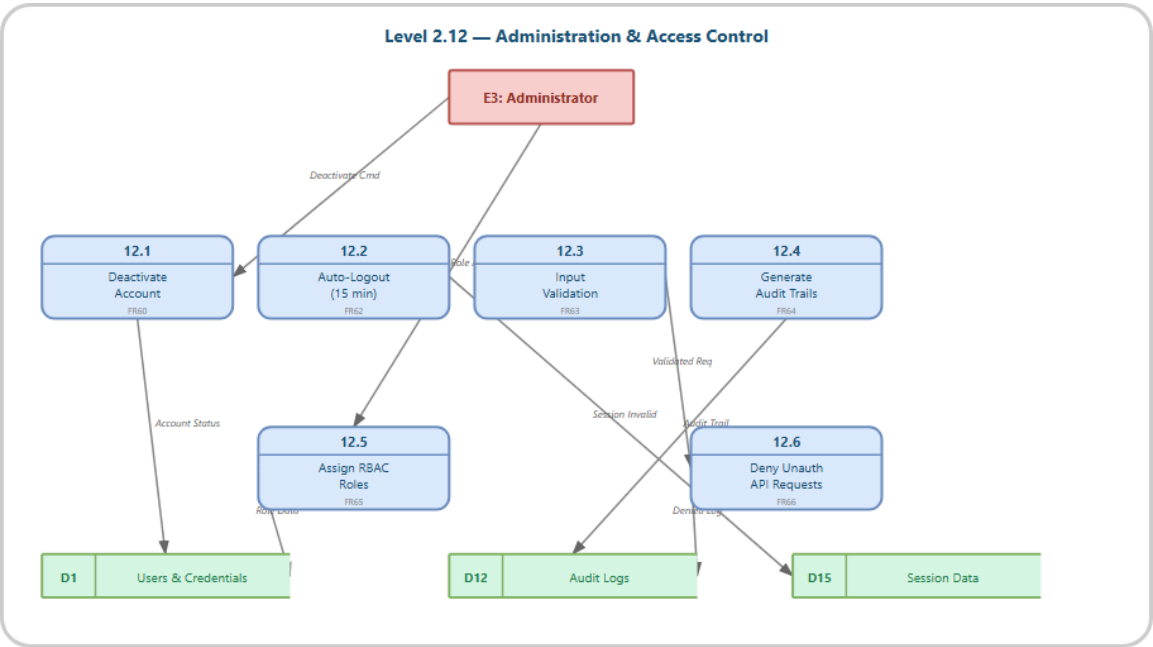

Figure 7.2.12: Level 2.12 — Administration & Access Control

Breaks down Process 13.0 into 6 steps: running a daily backup at midnight (FR45), restoring backups (FR46), logging system errors (FR47), alerting admins about critical issues (FR48), capping file uploads at 5 MB (FR73), and scheduling monthly maintenance (FR75).

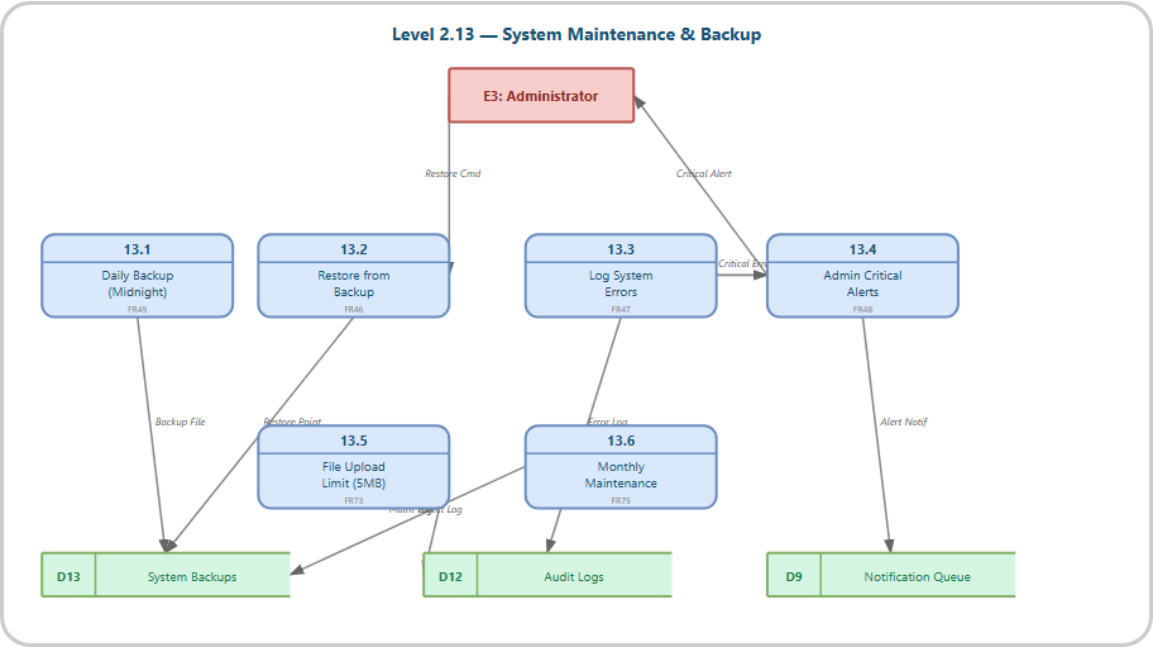

Figure 7.2.13: Level 2.13 — System Maintenance & Backup

Breaks down Process 14.0 into 3 steps: letting users revoke consent instantly (FR54), cutting off access as soon as consent is withdrawn (FR55), and logging every consent change (FR56).

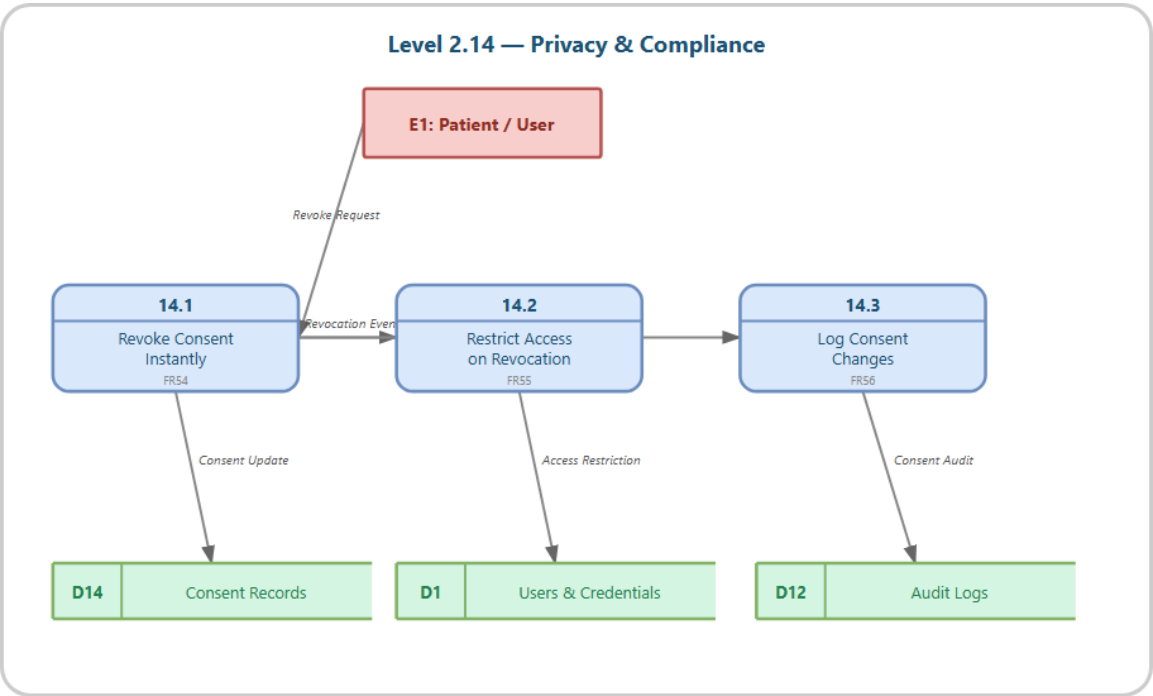

Figure 7.2.14: Level 2.14 — Privacy & Compliance

Breaks down Process 15.0 into 2 steps: choosing an interface language (FR49) and making sure the dashboard loads in under 3 seconds (FR50).

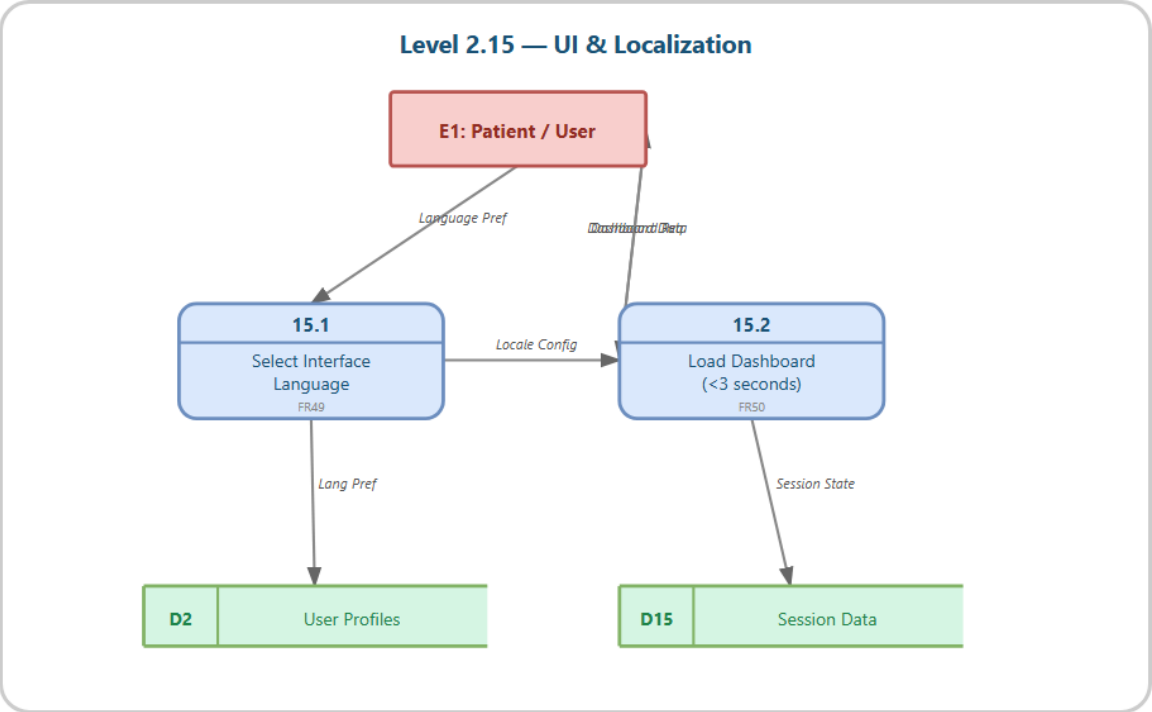

Figure 7.2.15: Level 2.15 — UI & Localization

8. Sequence Diagrams

The sequence diagrams show how different parts of the system talk to each other during key actions. They map out the order of messages between actors, the front end, the API server, the database, and any outside services.

This diagram walks through the full sign-up process: the user submits their details, a record is created in the database, an OTP is sent by email, the user enters the code, their account gets verified, a JWT token is issued, and the dashboard loads. The participants are the Patient, Frontend, API Server, and Database.

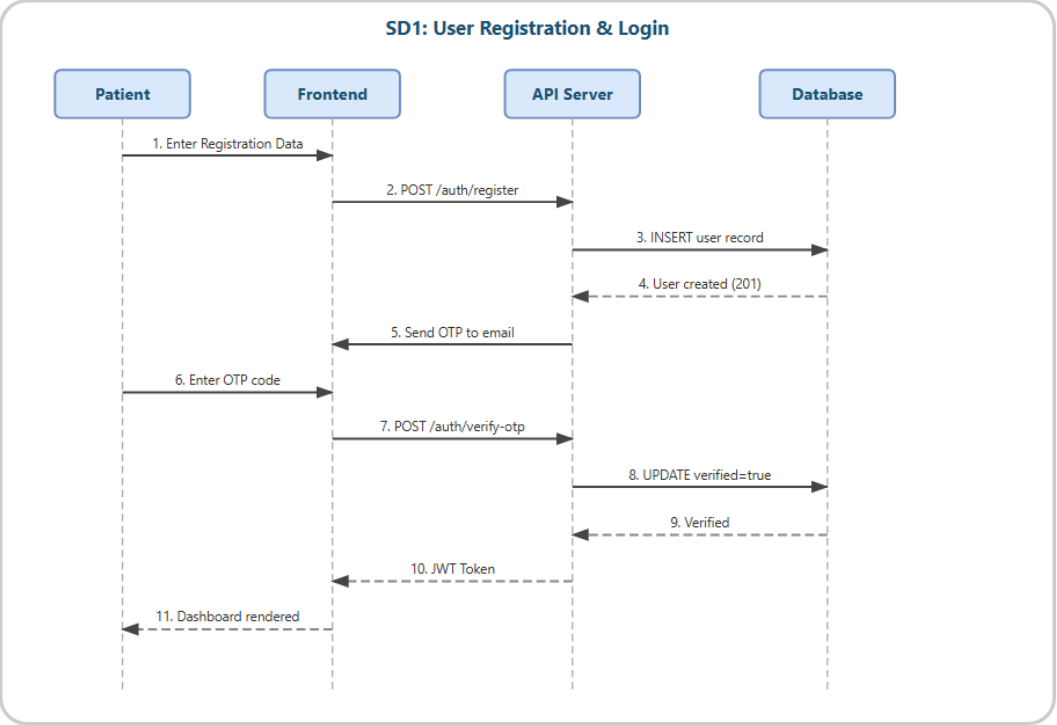

Figure 8.1: SD1 — User Registration & Login

This one covers appointment booking from start to finish: the user picks a clinician and date, the system checks if the slot is free, confirms the booking, saves it to the database, syncs with Google Calendar, and sends out an email notification. Participants include the Patient, Frontend, API Server, Database, and Google Calendar.

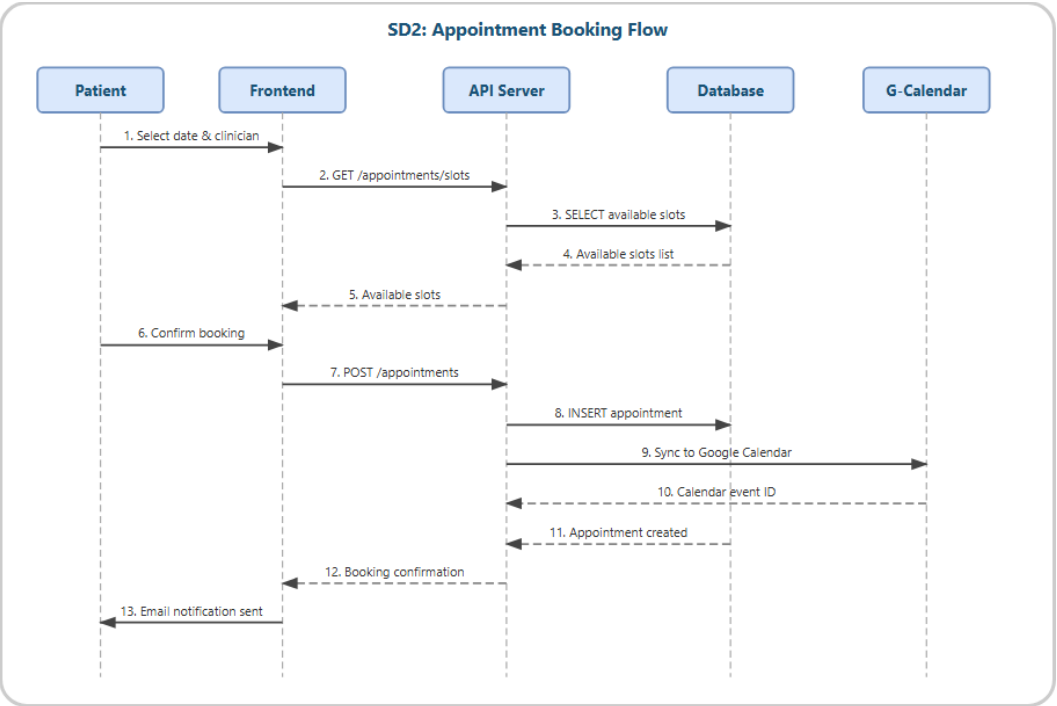

Figure 8.2: SD2 — Appointment Booking Flow

Here we show the mood submission flow. The system checks for duplicate entries within the last 24 hours, saves the mood score, looks for consecutive low scores (below 2 for 3 days straight), fires a risk alert if needed, logs a notification, and updates the dashboard.

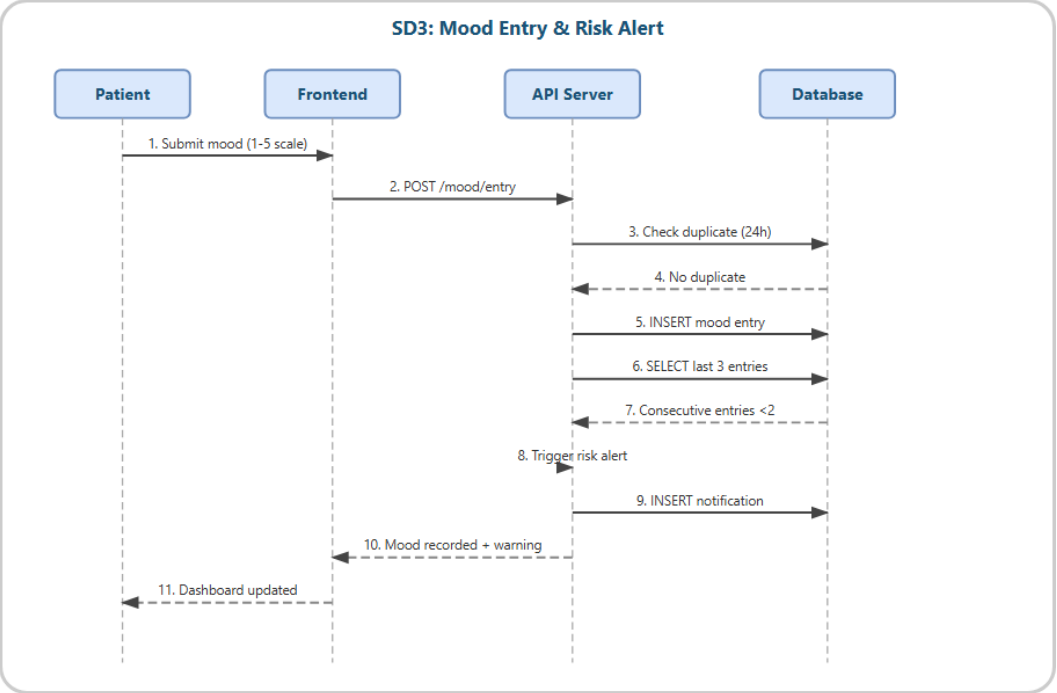

Figure 8.3: SD3 — Mood Entry & Risk Alert

This diagram covers a chatbot conversation: the user sends a message, the API server processes it, sends an NLP query to the AI engine, gets a confidence score back, delivers the response in under 3 seconds, saves the transcript, and — if confidence is below 70% — automatically passes the conversation to a real clinician.

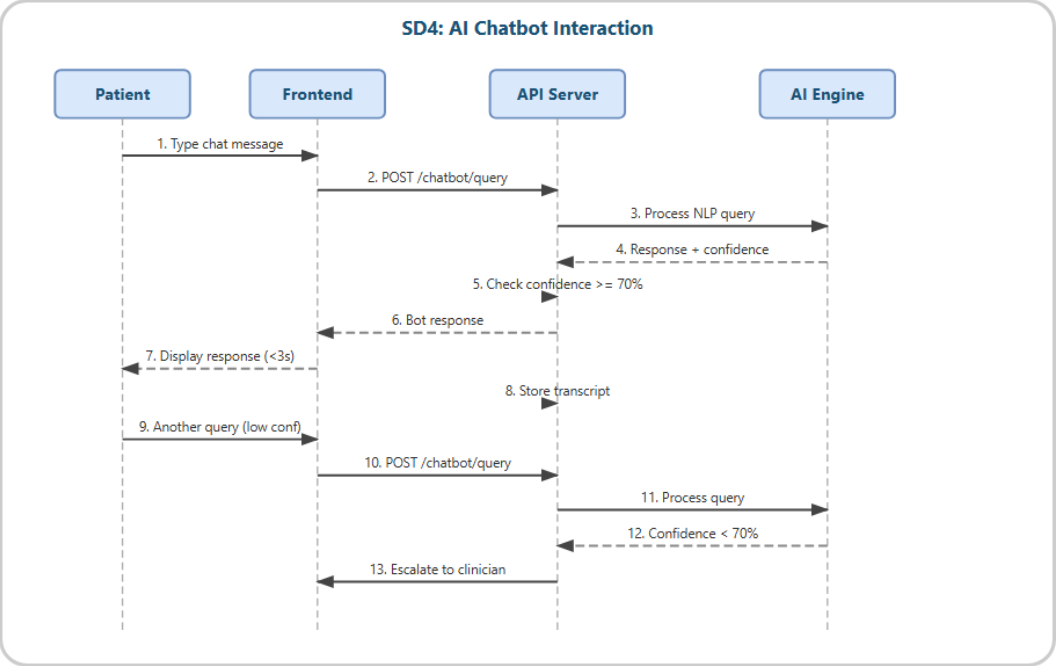

Figure 8.4: SD4 — AI Chatbot Interaction

This shows what happens when someone hits the emergency button: the system calls the emergency API, creates a log entry, fetches crisis hotline data from an outside service, displays the information in under 1 second, and records everything in the audit trail. Participants are the Patient, Frontend, API Server, Database, and Crisis Service.

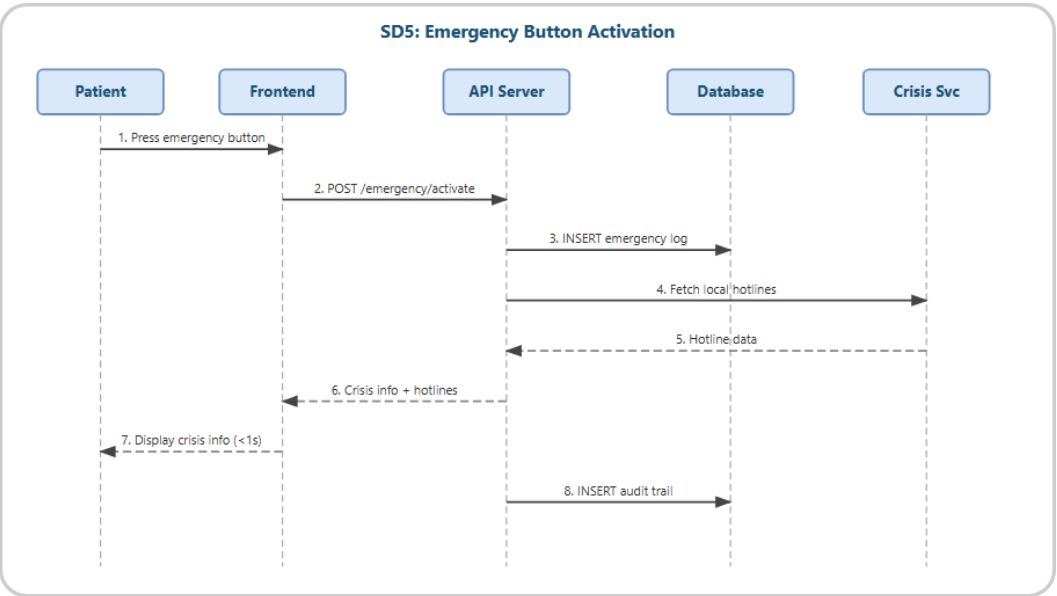

Figure 8.5: SD5 — Emergency Button Activation

This diagram follows a secure message from start to finish: the user writes the message, the server encrypts it with AES-256, stores the encrypted version, pushes it to the recipient in real time via WebSocket (under 2 seconds), confirms delivery, sends a notification, and supports soft deletion.

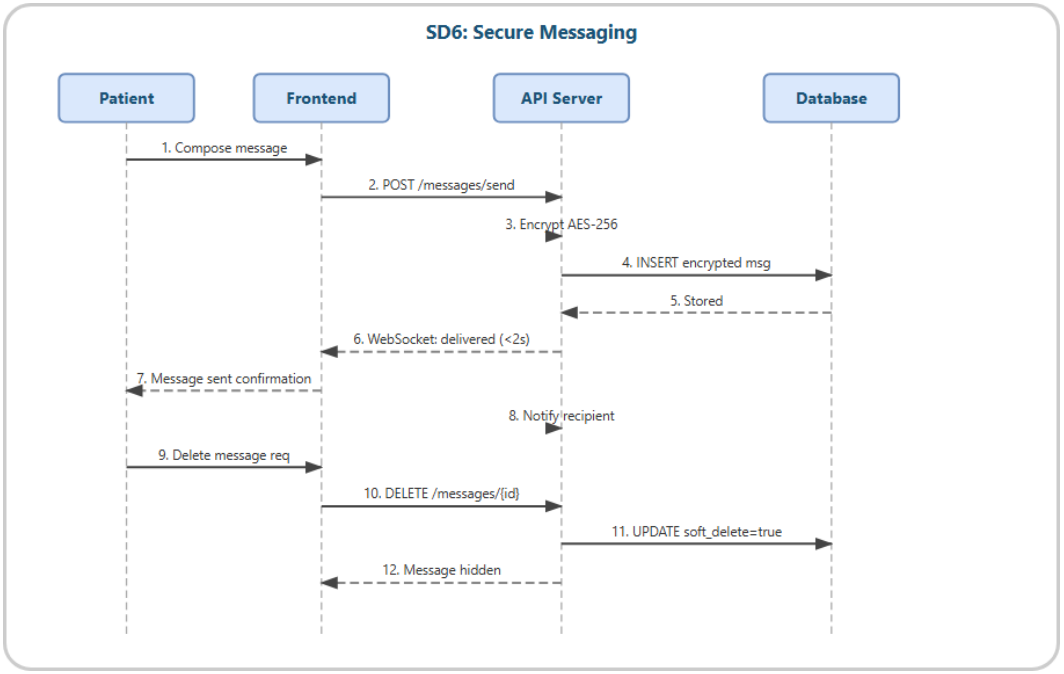

Figure 8.6: SD6 — Secure Messaging

9. Entity Relationship Diagram & Data Dictionary

The Entity Relationship Diagram (ERD) shows the logical data model behind our database. It lays out each table, its primary keys (PK) and foreign keys (FK), and how the tables relate to each other with clear cardinality.

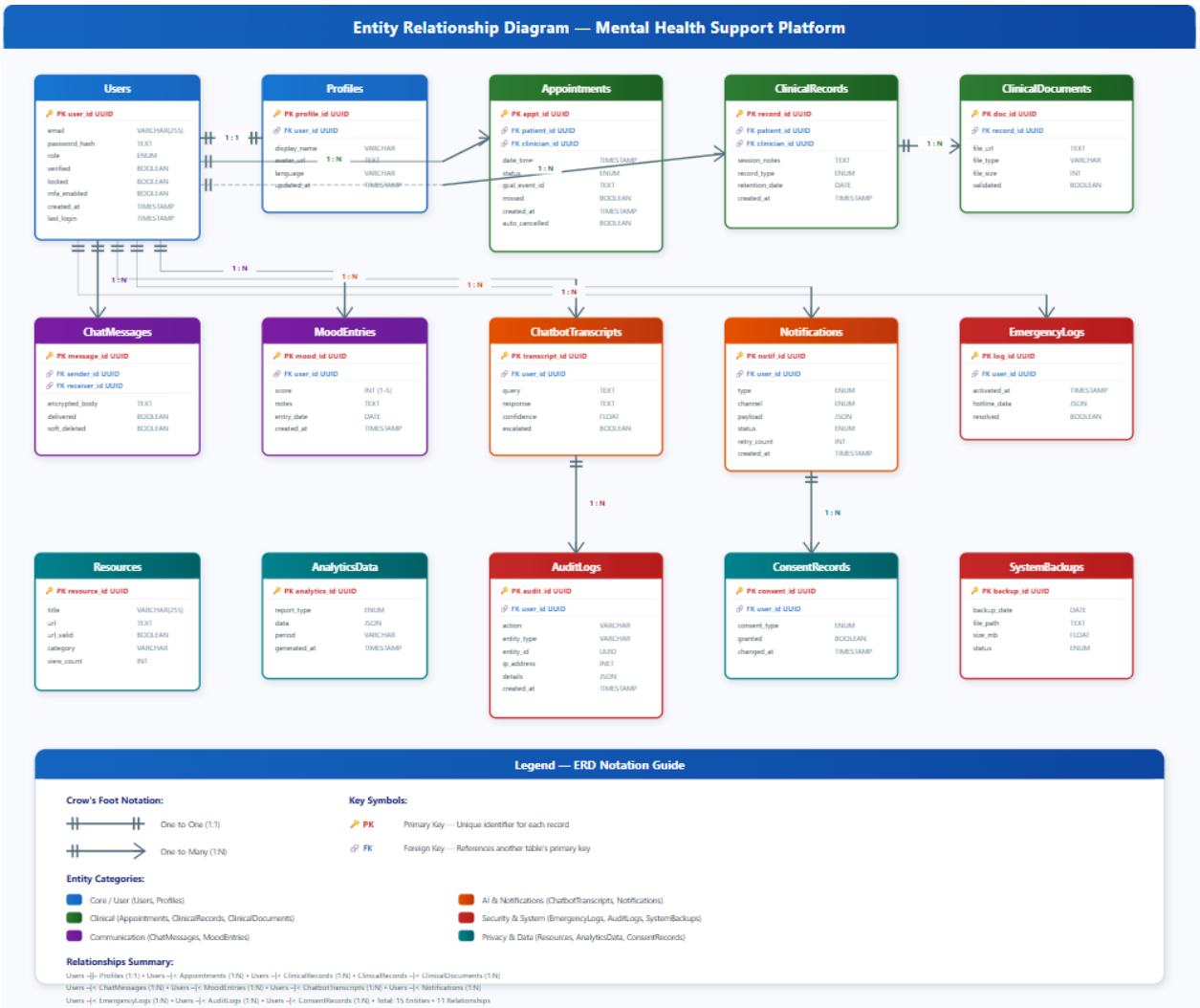

Figure 9.1: Entity Relationship Diagram — Mental Health Support Platform

9.1 Data Dictionary

Below are the schema definitions for every entity in the database:

Table: Users

Column	Type	Constraints	Description
user_id	UUID	PK	Unique user identifier
email	VARCHAR(255)	UNIQUE, NOT NULL	User email address
password_hash	TEXT	NOT NULL	bcrypt hashed password (10 salt rounds)
role	ENUM	NOT NULL	patient | clinician | admin
verified	BOOLEAN	DEFAULT false	Email/OTP verification status
locked	BOOLEAN	DEFAULT false	Account lock after 5 failed attempts
mfa_enabled	BOOLEAN	DEFAULT false	Multi-factor authentication flag

created_at	TIMESTAMP	DEFAULT NOW()	Account creation timestamp
last_login	TIMESTAMP	NULLABLE	Most recent login timestamp

Table: Profiles

Column	Type	Constraints	Description
profile_id	UUID	PK	Unique profile identifier
user_id	UUID	FK → Users	References Users table
display_name	VARCHAR(100)	NOT NULL	User display name
avatar_url	TEXT	NULLABLE	Profile image URL (< 2MB)
language	VARCHAR(10)	DEFAULT 'en'	Interface language preference
updated_at	TIMESTAMP	DEFAULT NOW()	Last profile update

Table: Appointments

Column	Type	Constraints	Description
appt_id	UUID	PK	Unique appointment identifier
patient_id	UUID	FK → Users	Patient reference
clinician_id	UUID	FK → Users	Clinician reference
date_time	TIMESTAMP	NOT NULL	Appointment date and time
status	ENUM	NOT NULL	pending | confirmed | missed | cancelled
gcal_event_id	TEXT	NULLABLE	Google Calendar event ID
missed	BOOLEAN	DEFAULT false	Missed after 15-min delay
created_at	TIMESTAMP	DEFAULT NOW()	Booking creation time
auto_cancelled	BOOLEAN	DEFAULT false	Auto-cancelled after 30 min

Table: ClinicalRecords

Column	Type	Constraints	Description
record_id	UUID	PK	Unique record identifier
patient_id	UUID	FK → Users	Patient reference
clinician_id	UUID	FK → Users	Clinician reference
session_notes	TEXT	NOT NULL	Clinical session notes
record_type	ENUM	NOT NULL	consultation | follow-up | emergency
retention_date	DATE	NULLABLE	Data retention deadline
created_at	TIMESTAMP	DEFAULT NOW()	Record creation time

Table: ChatMessages

Column	Type	Constraints	Description
message_id	UUID	PK	Unique message identifier
sender_id	UUID	FK → Users	Message sender
receiver_id	UUID	FK → Users	Message receiver
encrypted_body	TEXT	NOT NULL	AES-256 encrypted message body
delivered	BOOLEAN	DEFAULT false	Delivery confirmation
soft_deleted	BOOLEAN	DEFAULT false	Soft delete flag (not purged)

Table: MoodEntries

Column	Type	Constraints	Description
mood_id	UUID	PK	Unique mood entry identifier
user_id	UUID	FK → Users	User reference
score	INT	CHECK (1-5)	Mood score (1=very low, 5=excellent)
notes	TEXT	NULLABLE	Optional mood notes

entry_date	DATE	NOT NULL	Date of mood entry
created_at	TIMESTAMP	DEFAULT NOW()	Entry creation time

Table: ChatbotTranscripts

Column	Type	Constraints	Description
transcript_id	UUID	PK	Unique transcript identifier
user_id	UUID	FK → Users	User reference
query	TEXT	NOT NULL	User query text
response	TEXT	NOT NULL	AI-generated response
confidence	FLOAT	NOT NULL	AI confidence score (0.0–1.0)
escalated	BOOLEAN	DEFAULT false	Escalated to human (< 70%)

Table: Notifications

Column	Type	Constraints	Description
notif_id	UUID	PK	Unique notification identifier
user_id	UUID	FK → Users	Recipient user reference
type	ENUM	NOT NULL	booking | wellness | alert | follow_up
channel	ENUM	NOT NULL	email | sms | in_app
payload	JSON	NOT NULL	Notification content payload
status	ENUM	DEFAULT 'pending'	pending | sent | failed | queued
retry_count	INT	DEFAULT 0	API retry count (max 3)
created_at	TIMESTAMP	DEFAULT NOW()	Notification creation time

Table: EmergencyLogs

Column	Type	Constraints	Description
log_id	UUID	PK	Unique emergency log identifier
user_id	UUID	FK → Users	User who activated emergency
activated_at	TIMESTAMP	DEFAULT NOW()	Activation timestamp
hotline_data	JSON	NULLABLE	Crisis hotline information
resolved	BOOLEAN	DEFAULT false	Resolution status

Table: ConsentRecords

Column	Type	Constraints	Description
consent_id	UUID	PK	Unique consent identifier
user_id	UUID	FK → Users	User reference
consent_type	ENUM	NOT NULL	data_sharing | analytics | marketing
granted	BOOLEAN	NOT NULL	Consent granted or revoked
changed_at	TIMESTAMP	DEFAULT NOW()	Last consent change time

Table: AuditLogs

Column	Type	Constraints	Description
audit_id	UUID	PK	Unique audit entry identifier
user_id	UUID	FK → Users	User who performed action
action	VARCHAR(100)	NOT NULL	Action performed
entity_type	VARCHAR(50)	NOT NULL	Affected entity type
entity_id	UUID	NULLABLE	Affected entity ID
ip_address	INET	NULLABLE	Client IP address
details	JSON	NULLABLE	Additional action details
created_at	TIMESTAMP	DEFAULT NOW()	Audit entry time

10. Class Diagram

The UML Class Diagram maps out the object-oriented structure of the platform. Each class lists its attributes (properties) and methods (what it can do), with lines showing how the classes are linked through associations, dependencies, and multiplicities.

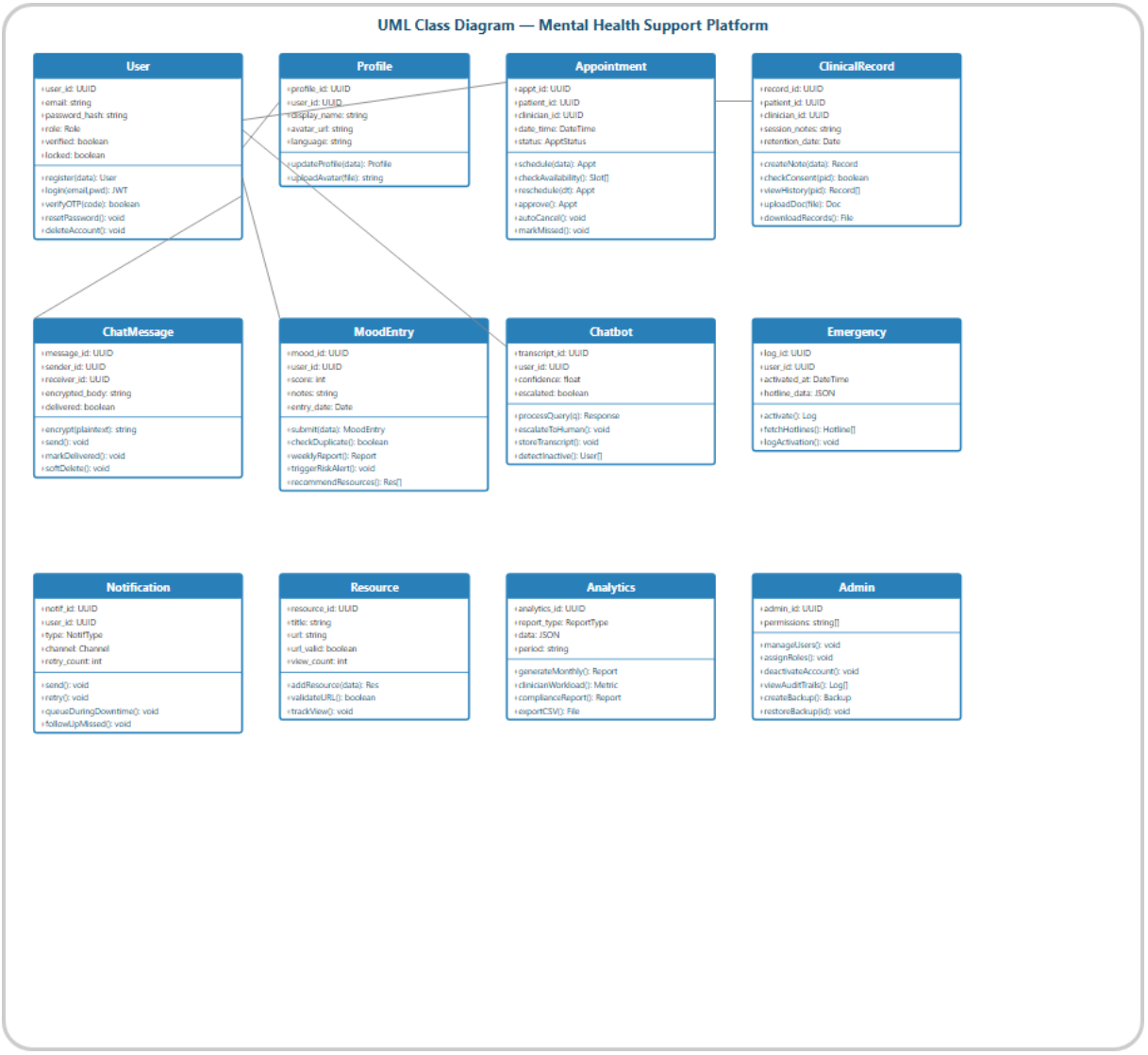

Figure 10.1: UML Class Diagram — Mental Health Support Platform

The system comprises 12 primary classes:

- User — Handles sign-up, login, OTP checks, password resets, and account deletion
- Profile — Manages the user's display name, profile picture, and language setting
- Appointment — Covers scheduling, clash detection, slot checking, rescheduling, approvals, auto-cancel, and missed marking
- ClinicalRecord — Deals with medical records, consent checks, session notes, document uploads and downloads, and viewing patient history

- ChatMessage — Sends and receives encrypted messages (AES-256) with delivery tracking and soft delete
- MoodEntry — Tracks daily moods, stops duplicate entries, builds weekly reports, raises risk alerts, and suggests resources
- Chatbot — Runs AI conversations with NLP, can escalate to a human, stores transcripts, and spots inactive users
- Emergency — Activates the emergency feature, pulls up hotline data, and logs each activation
- Notification — Sends alerts through email, SMS, and in-app channels with retry logic and downtime queuing
- Resource — Manages the mental health resource library, checks URLs, and tracks view counts
- Analytics — Produces monthly reports, tracks clinician workloads, generates compliance reports, and exports CSV files
- Admin — Gives admins control over user management, role assignments, account deactivation, audit logs, and backups

11. Data Protection Impact Assessment (DPIA)

The Data Protection Impact Assessment (DPIA) looks at the privacy risks that come with handling personal and sensitive health information on our platform, and explains the steps we take to reduce those risks.

11.1 Overview of Data Processing

The platform processes several categories of personal data:

- Account Data — Things like email addresses, hashed passwords, login times, and IP addresses
- Profile Data — Display names, profile pictures, and language choices
- Health Data — Mood scores, mood notes, clinical session notes, and medical records
- Communication Data — Encrypted chat messages and chatbot conversation logs
- Behavioural Data — How long sessions last, usage patterns, and resource view counts
- Emergency Data — Logs from emergency activations and crisis interactions
- Consent Data — What data-sharing choices the user has made and any changes over time

11.2 Risk Assessment

Risk	Severity	Mitigation Measures	Residual
Unauthorized access to health data	High	RBAC, JWT auth, consent-based access (FR17, FR65)	Low
Data breach of personal information	High	AES-256 encryption, bcrypt hashing, HTTPS (FR23, NFRS03)	Low
Unintended data retention	Medium	Right to deletion (FR8d), retention policies (FR19)	Low
Unauthorized data sharing	High	Consent management (FR54–55), immediate revocation	Low
Session hijacking	Medium	JWT 1hr expiry, auto-logout 15min (FR62, NFRS02)	Low
SQL injection / XSS attacks	High	Input validation (FR63), parameterized queries (NFRS05)	Low
Loss of data	Medium	Daily automated backups (FR45), restore within 1hr (FR46)	Low
Excessive API access	Medium	Rate limiting 100 req/IP/min (NFRS06)	Low
Emergency data misuse	Medium	Audit logging (FR40), access logging (FR22)	Low
Non-compliance with privacy laws	High	GDPR alignment, consent management (NFRC01–04)	Low

11.3 Privacy Controls Summary

Control	Implementation
Encryption at Rest	AES-256 for all sensitive data including messages and clinical records
Encryption in Transit	HTTPS/TLS 1.2+ for all client-server communications
Password Security	bcrypt hashing with 10 salt rounds; no plaintext storage
Authentication	JWT tokens with 1-hour expiry; OTP verification; multi-factor authentication
Access Control	Role-Based Access Control (RBAC) with Patient, Clinician, and Admin roles
Consent Management	Granular consent with instant revocation and immediate access restriction
Audit Trails	Complete logging of all database transactions, record access, and consent changes
Data Minimization	Only essential data collected; data anonymisation for analytics
Right to Deletion	User account deletion with data anonymisation within 24 hours
Backup & Recovery	Daily automated backups at midnight with 7-day retention; 1-hour restore SLA

12. Test and Implementation Plan

12.1 Testing Strategy

Our testing strategy uses multiple layers to make sure every part of the system works properly:

Test Type	Tool	Scope	Acceptance Criteria
Unit Testing	Jest	Individual functions, API endpoints, React components	≥ 80% code coverage
Integration Testing	Jest + Supertest	API endpoint chains, database interactions, auth flows	All critical paths
End-to-End Testing	Cypress	Complete user workflows from UI to database	Core user journeys
Security Testing	OWASP ZAP / Manual	SQL injection, XSS, auth bypass, rate limiting	Zero critical vulnerabilities
Performance Testing	Artillery / k6	Page load < 3s, API response < 500ms, 50 concurrent users	Meet all NFR targets
User Acceptance Testing	Manual	Stakeholder validation of all features	All FRs verified

12.2 Test Cases Summary

ID	Test Case	FR	Procedure / Expected Result	Status
TC001	User Registration	FR1	Register with valid email → Account created	Pass
TC002	Duplicate Email Rejection	FR1	Register with existing email → Error message	Pass
TC003	OTP Verification	FR2	Enter valid OTP within 5s → Account verified	Pass
TC004	Account Lockout	FR3	5 failed logins → Account locked	Pass
TC005	Password Reset	FR5	Reset with valid token (< 10 min) → Password updated	Pass
TC006	Mood Entry Submission	FR27	Submit mood 1–10 → Entry recorded	Pass
TC007	Duplicate Mood Prevention	FR28	Submit twice within 1 hour → Rejected	Pass
TC008	Risk Alert Trigger	FR30	3 consecutive mood < 2 → Alert generated	Pass
TC009	Appointment Booking	FR9	Book available slot → Appointment created	Pass
TC010	Conflict Detection	FR10	Book occupied slot → Rejected	Pass
TC011	Emergency Button	FR38	Press emergency → Crisis info < 1s	Pass
TC012	Message Encryption	FR23	Send message → Stored encrypted (AES-256)	Pass
TC013	Chatbot Response	FR32	Send query → Response < 3 seconds	Pass
TC014	Chatbot Escalation	FR33	Low confidence → Escalated to clinician	Pass
TC015	Consent Revocation	FR54	Revoke consent → Access immediately restricted	Pass
TC016	Auto-Logout	FR62	15 min inactivity → Session terminated	Pass
TC017	RBAC Enforcement	FR65	Patient access admin route → 403 Forbidden	Pass
TC018	SQL Injection Prevention	FR63	Inject SQL in input → Sanitized/blocked	Pass
TC019	File Upload Limit	FR73	Upload > 5MB file → Rejected	Pass
TC020	Dashboard Load Time	FR50	Load dashboard → Render < 3 seconds	Pass

12.3 Implementation Plan

Phase	Focus	Details
Phase 1 Week 7	Backend Foundation	Express.js setup, PostgreSQL schema migration, JWT + OTP authentication, user CRUD APIs
Phase 2 Week 8	Core Feature APIs	Mood tracking, appointment scheduling, messaging (AES-256), chatbot integration (OpenAI), emergency support APIs
Phase 3 Week 9	Frontend Development	React component library, dashboard, mood tracker UI, messaging UI, appointment booking UI, admin panel
Phase 4 Week 10	Integration & Real-time	Frontend-backend integration, Socket.io real-time chat, Google Calendar sync, notification engine, end-to-end testing
Phase 5 Week 11	Testing & Deployment	Unit/integration/E2E testing, security audit, performance testing, cloud deployment (Render/AWS), CI/CD pipeline
Phase 6 Week 12	Demo & Handover	Final demonstration, documentation completion, source code handover, deployment documentation

13. Responses on Feedback from Supervisor

In this section, we document how the team responded to feedback from our academic supervisor, Mr. Syed Altaf, and our industry sponsor, Nabin Singh, during the weekly project review meetings.

Week	Feedback	Response / Action Taken
Week 2	Clarify the scope boundaries — what is in/out of scope for the prototype	Added explicit scope and limitations section (Section 1.3) distinguishing prototype features from production-level features. Clearly stated that video consultation, mobile-native app, and real clinical diagnosis are out of scope.
Week 3	Ensure functional requirements are measurable and testable	Revised all 75 FRs to include specific measurable criteria (e.g., OTP within 5s, mood duplicate within 1hr, emergency response < 1s, API response < 500ms). Each FR now has a corresponding test case.
Week 4	Architecture should clearly show separation of concerns	Designed and documented three-tier architecture (Presentation, Application, Data) with clear component separation. Added detailed architecture diagram with technology labels for each layer.
Week 5	Data Flow Diagrams should follow Gane & Sarson method consistently	Redesigned all DFDs using Gane & Sarson notation. Created Level 0 Context Diagram, Level 1 DFD with 15 processes, and 15 Level 2 DFDs decomposing each process with proper data store references.
Week 5	ERD should be database-ready with proper normalization	Designed ERD with 15 normalized entities, UUIDs as primary keys, proper foreign key relationships, and cross-reference modulation. Created comprehensive data dictionary with constraints.
Week 6	Include security considerations throughout the design	Integrated DPIA (Section 11), implemented privacy controls table, added security-specific NFRs (NFRS01-06), and ensured all security FRs (FR23, FR54-55, FR62-66) are covered in DFDs and test cases.

References

[1] Sommerville, I. (2016). Software Engineering (10th ed.). Pearson Education.

[2] Pressman, R. S. & Maxim, B. R. (2020). Software Engineering: A Practitioner's Approach (9th ed.). McGraw-Hill.

[3] IEEE (1998). IEEE Std 830-1998: Recommended Practice for Software Requirements Specifications. IEEE.

[4] Gane, C. & Sarson, T. (1979). Structured Systems Analysis: Tools and Techniques. Prentice-Hall.

[5] OWASP Foundation (2021). OWASP Top 10 — 2021. https://owasp.org/Top10/

[6] React.js Documentation (2024). React — A JavaScript Library for Building User Interfaces. https://react.dev/

[7] Express.js Documentation (2024). Express — Fast, Unopinionated, Minimalist Web Framework. https://expressjs.com/

[8] PostgreSQL Documentation (2024). PostgreSQL: The World's Most Advanced Open Source Database. https://www.postgresql.org/docs/

[9] OpenAI API Documentation (2024). API Reference. https://platform.openai.com/docs/

[10] Google Calendar API Documentation (2024). Calendar API Overview. https://developers.google.com/calendar/

[11] Twilio Documentation (2024). Programmable Messaging. https://www.twilio.com/docs/

[12] SendGrid Documentation (2024). Email API. https://docs.sendgrid.com/

[13] Socket.IO Documentation (2024). Socket.IO. https://socket.io/docs/

[14] W3C (2018). Web Content Accessibility Guidelines (WCAG) 2.1. https://www.w3.org/TR/WCAG21/

[15] Information Commissioner's Office (ICO) (2022). Data Protection Impact Assessments. https://ico.org.uk/for-organisations/guide-to-data-protection/

Appendix

A. Table of Figures

Figure #	Description
Figure 1.1	Organisation Chart — Mel23 Tech Solution
Figure 1.2	Work Breakdown Structure
Figure 3.1	Three-Tier System Architecture Diagram
Figure 4.1	Login / Registration Page Wireframe
Figure 4.2	User Dashboard Wireframe
Figure 4.3	Mood Tracker Page Wireframe
Figure 4.4	Secure Messaging Interface Wireframe
Figure 4.5	Appointments Page Wireframe
Figure 4.6	Admin Dashboard Panel Wireframe
Figure 5.1	Patient / User Use Case Diagram
Figure 5.2	Clinician / Doctor Use Case Diagram
Figure 5.3	Administrator Use Case Diagram
Figure 6.1	Level 0 Context Diagram (Gane & Sarson)
Figure 7.1	Level 1 System Data Flow Diagram
Figure 7.2.1	Level 2.1 — Authentication & Account Management DFD
Figure 7.2.2	Level 2.2 — Profile Management DFD
Figure 7.2.3	Level 2.3 — Appointment Management DFD
Figure 7.2.4	Level 2.4 — Clinical Records Management DFD
Figure 7.2.5	Level 2.5 — Messaging System DFD
Figure 7.2.6	Level 2.6 — Mood Tracking & Analysis DFD
Figure 7.2.7	Level 2.7 — AI Chatbot System DFD
Figure 7.2.8	Level 2.8 — Emergency Support DFD
Figure 7.2.9	Level 2.9 — Notification Engine DFD
Figure 7.2.10	Level 2.10 — Content & Resource Management DFD
Figure 7.2.11	Level 2.11 — Analytics & Reporting DFD
Figure 7.2.12	Level 2.12 — Administration & Access Control DFD
Figure 7.2.13	Level 2.13 — System Maintenance & Backup DFD
Figure 7.2.14	Level 2.14 — Privacy & Compliance DFD
Figure 7.2.15	Level 2.15 — UI & Localization DFD
Figure 8.1	SD1 — User Registration & Login Sequence Diagram
Figure 8.2	SD2 — Appointment Booking Flow Sequence Diagram
Figure 8.3	SD3 — Mood Entry & Risk Alert Sequence Diagram
Figure 8.4	SD4 — AI Chatbot Interaction Sequence Diagram
Figure 8.5	SD5 — Emergency Button Activation Sequence Diagram
Figure 8.6	SD6 — Secure Messaging Sequence Diagram
Figure 9.1	Entity Relationship Diagram
Figure 10.1	UML Class Diagram

B. Table of Tables

Table #	Description
Table 1.1	Company Information
Table 1.2	Roles and Responsibilities
Table 1.3	Key Deliverables
Table 1.4	Project Budget Estimates
Table 1.5	WBS Phase Summary
Table 1.6	Gantt Chart / Project Timeline
Table 2.1	Functional Requirements (FR1–FR75)
Table 2.2	Non-Functional Requirements
Table 3.1	Technology Stack
Table 7.1	Level 1 Process Descriptions
Table 7.2	Data Store Definitions
Table 9.1–9.11	Data Dictionary Tables
Table 11.1	DPIA Risk Assessment
Table 11.2	Privacy Controls Summary
Table 12.1	Testing Strategy

Table 12.2	Test Cases Summary
Table 12.3	Implementation Plan
Table 13.1	Supervisor Feedback Responses